

Recent Advancements in Topic Modeling Techniques for Healthcare, Bioinformatics, and Other Potential Applications

Pratima Kumari, Sachin Kadian, Mukund Vora, Pradyumna Vemuri, Sumit Kumar, Om Prakash Verma, and Roger J. Narayan*

In today's era of enormous text data, topic modeling is emerging as a revolutionary tool in natural language processing. From the corpus of scientific research articles to social media posts and newspaper headlines, topic modeling is employed in several domains to discover the latent primary themes associated with the corpus. This article provides an extensive and comprehensive review of different topic modeling techniques from their origin to the present. The effectiveness and efficacy of different topic modeling techniques, such as non-negative matrix factorization, latent Dirichlet allocation, latent semantic analysis, probabilistic latent semantic analysis, Top2Vec, and BERTopic, are reviewed to highlight their strengths and weaknesses. A concise summary of recent studies in healthcare, bioinformatics, scientific research articles, social media platforms, and legal domains is also presented. Different quantitative and qualitative evaluation metrics are also discussed to understand the performance of topic modeling techniques better. Finally, a brief discussion on existing challenges and prospects of topic modeling is also included, providing researchers with insight into further advancements in topic modeling.

and understand the context; however, additional efforts and time are required to discover the relevant information about a specific topic of interest from the vast amount of available data and sources of knowledge.^[2] The knowledge of latent context associated with available data can help enhance the performance of different natural language processing applications, including summarization, document classification, text extraction, and dimensionality reduction.^[3–6] Therefore, there is a demand for effective automatic tools to analyze the available data and recognize the underlying core topics or common contexts associated with the data.^[7,8] To automate the task of understanding the topics associated with a collection of documents/data, researchers utilize a technique known as topic modeling.^[9,10]

Topic modeling can be described as an unsupervised learning technique that can be employed to find hidden topics that are associated with a large, unlabeled dataset.^[11,12] The documents in the dataset can be tagged with the identified topics, which assist end users and researchers to understand each document and the whole collection. Over the past few years, topic modeling has garnered significant attention among researchers related to its use in various research fields, such as text mining, computer vision, recommendation systems, and genetics.^[13,14]

1. Introduction

With the emergence of the World Wide Web, an enormous amount of information is stored on the Internet. This information is in the format of text data, which can be transcribed from human-generated social media posts, digital libraries, documents, images, audio, and product reviews.^[1] Humans can read this available data

P. Kumari
 Center for Digital Health Innovation
 University of California
 San Francisco 94143, CA, USA

S. Kadian, R. J. Narayan
 Lampe Joint Department of Biomedical Engineering
 University of North Carolina and North Carolina State University
 Raleigh 27695, NC, USA
 E-mail: rjnaraya@ncsu.edu

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/aisy.202400528>.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

DOI: 10.1002/aisy.202400528

M. Vora, P. Vemuri
 Department of Computer Science
 North Carolina State University
 Raleigh 27695, NC, USA

S. Kumar
 Department of Electronics & Communication, Faculty of Engineering & Technology
 JAIN (Deemed-to-be University)
 Bangalore 562112, India

O. P. Verma
 Department of Instrumentation and Control Engineering
 Dr B R Ambedkar National Institute of Technology Jalandhar
 Jalandhar 144027, Punjab, India

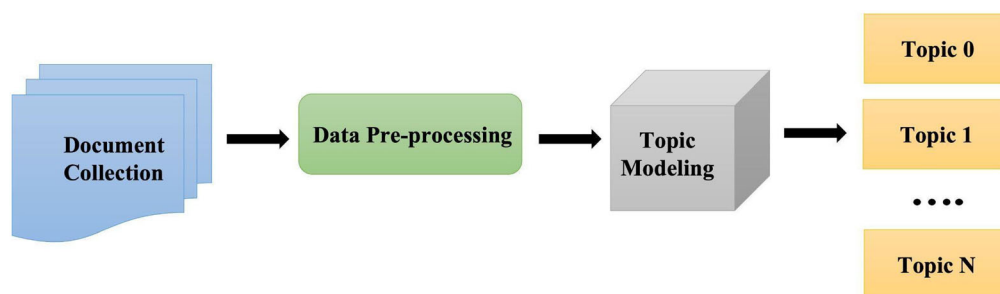


Figure 1. Schematic diagram of topic modeling.

A few examples of these fields are as follows: classifying databases of research articles and journals into groups based on similar topics,^[15–17] understanding the trends on social media platforms (e.g., X (previously referred to as Twitter) and Facebook) based on different public surveys,^[18–22] and scheduling different optimization plans using healthcare data.^[23–25] **Figure 1** shows the flow-chart of the topic modeling technique. Initially, documents are collected from a direct or indirect source based on the application. Direct sources mean the considered dataset is a collection of documents that contain text. Indirect sources can be the transcription of audio files, videos, and images into text format, which can be addressed as documents. Topic modeling considers that different hidden topics are associated with these documents.^[26] Although topic modeling cannot determine the exact meaning of each word in the document, it captures the context of the text to determine the underlying concepts by assigning probabilities to terms in the documents.^[27] Topic modeling is a technique that can be considered similar to clustering, which maps a collection of documents to a low-dimensional topic subspace. However, clustering and topic modeling are more closely related than they might initially appear. A nuanced perspective reveals that topic modeling often leverages clustering techniques to uncover deeper semantic structures within text.^[28] Probabilistic models such as probabilistic latent semantic analysis (pLSA) and latent Dirichlet allocation (LDA) represent documents as mixtures of topics, and topics as mixtures of words, emphasizing a soft clustering approach. In contrast, modern

methods like BERTopic and Top2Vec incorporate algorithms such as hierarchical density-based spatial clustering of applications (HDBSCAN) to assign documents to discrete topics, effectively mimicking hard clustering. Apart from finding topics, topic modeling gives a few other insights about the documents that can be utilized in further analysis and applications. The topic modeling technique gives three outputs when applied to the corpus of the documents, which are as follows: set of topics (all identified topics in a corpus), percentage of topics that are identified in each document (indicates the proportion of each topic present in an individual document), and frequency of words in each topic (shows how often each word appears within a topic), as shown in **Figure 2**. These outputs provide a provision to explain the corpus in multiple ways. For instance, it reveals the hidden topics in the corpus, explains the proportion of the corpus that falls under different categories based on their topics, describes the topic proportion of a document, and similar types of outputs.^[29–31]

Different types of topic modeling approaches have been proposed by researchers over the past two decades, which include statistical and probabilistic approaches, matrix factorization or linear algebra approaches, and NLP-aided embedding approaches. Latent semantic analysis (LSA)^[32] and non-negative matrix factorization (NMF)^[33] are some of the most recognized linear algebra approach-based topic modeling techniques. Similarly, pLSA^[34,35] and LDA are the two most popular techniques utilized for topic modeling that are based on the

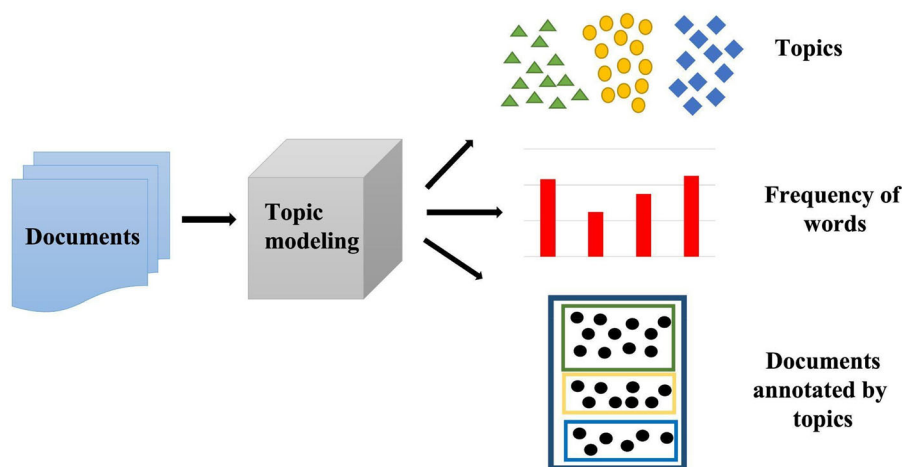


Figure 2. Topic modeling technique for information retrieval.

probabilistic approach. Moreover, several types of derivations of LDA have been proposed as per the application, which includes latent Dirichlet mixture model (LDMM),^[31,36] online LDA, matrix factorization through LDA (fLDA),^[37] labeled LDA (L-LDA), and the hierarchical LDA (hLDA).^[38] Each of the models is derived to overcome the drawback and enhance the performance by combining basic features of LDA with more advanced probabilistic or statistical processes. NLP-aided embedding-based approaches, which include BERTopic and Top2Vec, are the most recent and advanced topic modeling techniques. Numerous topic modeling algorithms have been proposed so far that have been applied in different applications and domains.^[39,40] These domains include several business areas such as hospitality, food, tour and travel, pharmaceuticals, medical, healthcare, banking, finance, and various social media platforms (e.g., X (Twitter) and Facebook).^[41–43] Despite their popularity, topic modeling techniques suffer from different issues with optimization, stability, and noise handling in data, which leads to less accurate performance. Moreover, there are a variety of document types and applications; as such, there is no single topic modeling approach that can perform best in every scenario.^[13,26,44] A number of factors influence the selection of appropriate topic modeling techniques for different applications.^[45,46]

This article discusses the most commonly used topic modeling techniques as well as the strengths, weaknesses, and challenges addressed by these techniques. Moreover, most review articles are mostly limited to linear algebra and probabilistic approach-based models, including latent semantic indexing (LSI), NMF, pLSA, and LDA. In addition to the aforementioned approaches, this review article also focuses on the most advanced and new NLP-aid embedded approach-based models, which include BERTopic, Top2Vec, and their variants. This article will comprehensively review different topic modeling techniques and their applicability in different domains. This approach will provide a deeper insight into technique which is better for a given scenario. A different number of factors, such as document type, length of text, size of documents, sparsity, and downstream tasks, help in choosing the most appropriate modeling techniques. Tools and packages for topic modeling are described for implementing the discussed models. To the best of our knowledge, no previous studies have provided a comprehensive review of all the latest existing topic modeling techniques. Section 2 elaborates on different types of topic modeling techniques that have evolved so far, along with the strengths, weaknesses,

and challenges addressed by these models. A detailed review of the applications utilizing topic modeling is discussed in Section 3. An explanation of evaluation metrics that are utilized to understand the performance of these existing models is provided in Section 4. Section 5 explains different challenges and deciding factors in topic modeling. Finally, Section 6 provides a discussion and considers future prospects involving topic modeling techniques.

2. Basic Approaches of Topic Modeling

2.1. LSA

LSA is the elementary topic modeling technique, which was introduced in 1990 by Deerwester et al.^[11] They explained that LSA automatically retrieves the documents from the database and assigns an index to them, named LSI. Moreover, LSI is a well-known information retrieval model, which creates vector-based representations of documents and claims to capture their semantic content. LSI assumes some type of underlying (or latent) structure in the word usage, which is partially obscured by word choice variability.

Further, singular value decomposition (SVD) is utilized for transforming the data into a lower-dimensional space, revealing underlying patterns in word usage across documents.^[47] SVD is a matrix algebra technique which is used to reorient and rank the dimensions within a vector space. SVD organizes the vectors in the order of most significant to least significant. Even if some of the dimensions that are less significant get ignored, this reduced representation is guaranteed to be the best representation that is available for that dimensionality.^[48–50] The statistically derived vectors serve as robust indicators of documents and help in performing the retrieval of documents from large databases.

2.1.1. Algorithm

Figure 3 demonstrates the steps followed in the LSA algorithm. The mathematical explanation of how the LSA algorithm performs on a group of documents is explained in the following steps: 1) Given m documents and n words in the vocabulary, the first step is to generate a $m \times n$ document-term matrix, referred to as A . In this approach, each row and column in this matrix represents a document and a word, respectively. 2) The

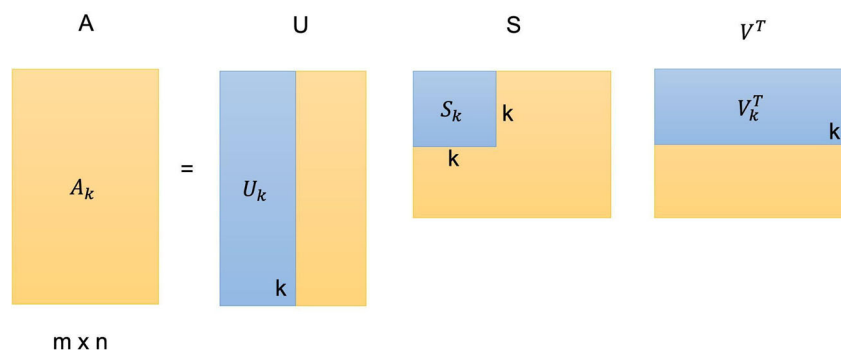


Figure 3. Singular value decomposition process.

entry corresponding to a document and word within the document-term matrix represents the *tf-idf* (term frequency-inverse document frequency) score. The higher the *tf-idf* score value, the more frequently a particular word is found across the document, but infrequently found across the corpus. 3) Generally, document-term matrix A is very noisy, sparse, and redundant over many dimensions. Therefore, in order to obtain the latent topics from a given collection of documents, a reduction of the dimensionality of the matrix A is necessary. The dimensionality reduction is undertaken by applying truncated SVD. The decomposition results in three matrices, represented as U , S , and V , as shown below:

$$A = USV^T \quad (1)$$

In this equation, U represents term eigenvectors, S contains a diagonal matrix of the singular values of A in descending order, V represents document eigenvectors, and T is the transpose operation. 4) The SVD reduces the dimensionality by keeping only the k highest singular values from V and discarding the remaining dimensions to produce a truncated version of the document-term matrix, A_k , as follows

$$A_k = U_k S_k V_k^T \quad (2)$$

In this equation, k represents the number of topics, which can be considered a hyperparameter that needs to be optimized. Smaller values of k can remove a significant amount of the noise; however, utilizing too few dimensions may also result in losing important information. Therefore, choosing the optimal value of k is an interesting problem.

2.1.2. Strengths and Weaknesses of LSA

LSA is one of the popular topic modeling techniques that is easy to implement, understand, as well as use. LSA yields more meaningful and diverse topics compared to the traditional vector space model. While the vector space model represents documents as high-dimensional vectors based solely on term frequencies, it fails to account for synonymy or latent semantic structure. This leads to sparse and overly high-dimensional representations, which LSA effectively mitigates by capturing underlying semantic relationships. Moreover, LSA is capable of handling synonymy (i.e., different terms or vocabulary are being used to describe the same underlying concept) to some extent. Unlike LSA, traditional methods rely on exact word matching and therefore struggle to retrieve documents that address the same topic but use different vocabulary. Similarly, LSA efficiently handles the issue of polysemy in the data (i.e., words having multiple meanings), which may significantly decrease the performance. A compressed representation of LSA may help to reduce the noise within the data.^[51,52]

Moreover, LSA provides comparatively faster runtime as it decomposes only the term-document matrix and achieves significant compression in large collections, which is better than many dimensionality reduction techniques. LSA is efficient and easy to implement; however, it has a few drawbacks. LSA does not consider word order while converting documents to the document-term matrix, which may lead to information loss in topic modeling, as word order is a significant parameter that

defines the meaning of words. It requires a very large corpus and vocabulary to achieve accurate results. In the case of LSA, finding an optimal number of topics is heuristic, and expertise is needed on the topic.

Moreover, it is a linear model and might not perform well on datasets with nonlinear dependencies. LSA uses SVD for dimensionality reduction, which works well with data following a Gaussian distribution. However, this distribution of data may not be possible in all cases. Finally, LSA suffers from a lack of interpretability, as it can be difficult to determine what topic these sets of words represent.^[51,53]

2.2. pLSI

In 1999, Thomas Hofmann described probabilistic latent semantic indexing (pLSI); this approach is an evolution from LSA; however, it adds a sounder probabilistic model.^[34] pLSA overcomes the drawbacks of the LSA model as it has a solid statistical foundation, unlike LSA. It is a statistical approach that is used in the analysis of two-mode as well as co-occurrence data. The key idea behind using pLSA is to determine a probabilistic model with the latent topics for obtaining data that is observed in the document-term matrix. pLSA employs the principle of log-likelihood estimation and defines a generative probabilistic model. Hoffman referred to topics as factors in his paper. He replaced the SVD with a probabilistic model and described it as an aspect model.

Furthermore, an expectation-maximization algorithm is utilized to train the model. LSA is able to map the words with a topic, while pLSA associates the word with a topic with a particular probability. In this way, a more meaningful model can be generated using pLSA, as each word is assigned to a topic that is determined from a multinomial distribution over topics. A probabilistic mixture of words makes a topic, and a mixture of multiple topics would make a document.^[54–56]

2.2.1. Algorithm

Figure 4 demonstrates the pictorial representation of the pLSA algorithm, which can be explained as follows. In pLSA, there are three sets of variables, which are defined as follows: First, documents, $D = \{d_1, d_2, d_3 \dots d_N\}$, in which d_i denotes the i_{th} document, and N denotes the total number of documents. Second, words, $W = \{w_1, w_2, w_3 \dots w_M\}$, in which w_j denotes the j_{th} word in the vocabulary with a size of M . Third, topics, $Z = \{z_1, z_2, z_3 \dots z_k\}$, in which k denotes the number of topics, which is specified by the user. The start point of the pLSA is the aspect model, which is a latent variable obtained from co-

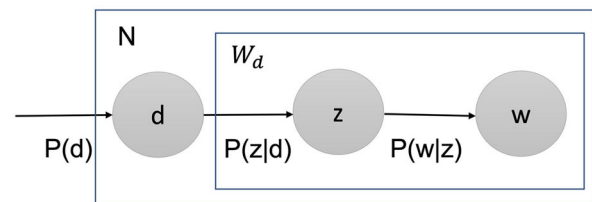


Figure 4. Graphical model of pLSA.

occurrence data (w, d) . A generative model to describe word/document co-occurrences can be developed using the following approach. 1) Choose a document d_i that is associated with probability $P(d_i)$. 2) Select a latent class z_k that is associated with probability $P(z_k)$. 3) Generate a word w_j that is associated with probability $P(z_k)$.

As such, the joint probability associated with a document and a word is defined as follows

$$P(d_i, w_j) = P(d_i)P(w_j|d_i) \quad (3)$$

where

$$P(d_i) = \sum_{k=1}^K P(w_j|z_k)P(z_k|d_i) \quad (4)$$

The equivalent symmetric version of the model is generated by inverting the conditional probability with the use of Bayes' rule.

$$P(d_i, w_j) = \sum_{k=1}^K P(z_k)P(d_i|z_k)P(w_j|z_k) \quad (5)$$

This equation is a reparameterized version of Equations (3) and (4).

Finally, the parameters associated with a pLSA model, $P(z_k)$ and $P(z_k|d_i)$, are estimated with an iterative expectation-maximization (EM) algorithm in order to fit a training corpus D through a maximization of the log-likelihood function L .

$$\sum_{d \in D} \sum_{w \in d} f(d_i, w_j) \log P(d_i, w_j) \quad (6)$$

In this equation, $f(d_i, w_j)$ describes the term frequency, which is the number of times that w occurred in d .

EM is a repetitive procedure that involves two phases for each iteration. In this approach, the first step is denoted as the Expectation step; the probabilities of the hidden classes (z) are determined based on the current parameter in this step. The second step is denoted as the Maximization step; in this step, the parameters are updated using information from the probabilities that are determined in the previous step. By alternating between the Expectation and Maximization steps, the algorithm reaches a converging point that represents a local maximum associated with the log-likelihood function, as shown in Equation (6).

2.2.2. Strengths and Weaknesses of pLSI

pLSA can deliver better performance than LSA as this approach uses a probabilistic model for words within the document, in which the topics can be viewed in terms of mixture component distributions. Moreover, the results of pLSA can be interpreted clearly.^[57,58] Although pLSA can overcome the drawbacks of LSA, it suffers from a few disadvantages as well. pLSA cannot perform well on unseen documents as it learns $P(z|d)$ only related to documents for which it is trained; as such, it is not considered a fully generative model. Moreover, pLSA is prone to scalability and overfitting since the number of parameters increases in a linear manner as the number of documents increases.^[59–61]

2.3. LDA

First described by Blei, Ng, and Jordan in 2003, LDA is the most well-known and frequently utilized of all the topic modeling techniques.^[62] LDA divides the data into three levels, which include topic, word, and document; this approach considers a document as a mixture of hidden topics, whereas each topic can be further represented by a set of words having a probability distribution. LDA assumes that each document's topic distribution is drawn from a Dirichlet prior, whereas the topic assignment of a word & word given by a topic are both multinomial distributions.^[63] Similar to pLSA, LDA also uses a bag-of-words (BoW) as well as a document-term matrix. The crucial difference between LDA and pLSA lies in the manner LDA models a document. Unlike pLSA, in which a single topic is assigned to each document, LDA views a document as a combination of multiple topics.^[31,64] In this way, it enables a document to be generated using the probabilistic distribution of many topics. LDA is driven by two key concepts: First, that every document is a combination of various topics, and second, that each topic is made up of a mixture of words. LDA develops a mathematical framework to reconcile these two principles simultaneously by determining the mixture of words that characterize each topic and determining the mixture of topics that are utilized to characterize each document.^[65,66]

2.3.1. Algorithm

The learning process of topic distribution in LDA is demonstrated through the plate notation in **Figure 5**. We consider a corpus $D = \{d_1, d_2, d_3, \dots, d_M\}$ that contains M documents. Each document contains a set of words that are described as $W_p = \{w_{p,1}, w_{p,2}, \dots, w_{p,N}\}$, where N is the number of words contained in document d_p and k describes the underlying latent topics. LDA models D using this generative process.

For each document $d \in D$: 1) Select a multinomial distribution θ_d for document d ($d \in \{1, \dots, M\}$) from a Dirichlet distribution with the parameter α . 2) Select a multinomial distribution ϕ_k for topic k ($k \in \{1, \dots, K\}$) from a Dirichlet distribution with the parameter β . 3) For a word w_i ($i \in \{1, 2, \dots, N\}$) in document d : a) Draw a topic z_i from θ_d . b) Draw a word w_i from ϕ_{z_i} .

In this equation, α , β , and k are three hyperparameters in LDA that need to be carefully tuned in order to achieve high

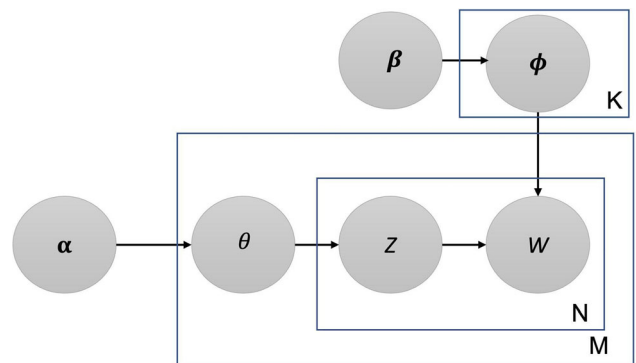


Figure 5. Schematic diagram of plate notation for LDA generative process.

performance; α is the document density factor, which controls the number of topics that are expected within the document. Setting α higher results in more topics per document. β is the topic word density factor, which controls the distribution of the words per topic in the document; a lower value of β results in fewer words per topic; k defines the number of topics that are able to be extracted from a corpus.

2.3.2. Strengths and Weaknesses of LDA

LDA is the most extensively utilized topic modeling technique; it has several advantages over other topic modeling techniques, including LSA and pLSA. One of the advantages of LDA over other models is that it understands unstructured data very well and provides better inference on unseen documents. Moreover, through LDA, words are disambiguated in a better approach, and the assignment of documents to the topics can be more precise. Also, the generative process of LDA can be utilized to generate synthetic documents that closely align with the statistical properties of the corpus.^[67–69] The principal advantages of LDA are that it provides scalability with an increase in the size of the corpus and handles the problem of overfitting. The LDA approach showed a substantial improvement over preceding techniques; however, it also suffers from several limitations. LDA is incapable of performing well on small documents. Another limitation associated with LDA is that it fails to capture the correlation between the topics. Also, LDA deals with optimizing the value of a few hyperparameters, such as α and β , which require expertise to maximize the marginal log-likelihood associated with the data. Moreover, the number of topics in the LDA is fixed. It must be known ahead of time; this parameter is one of the most significant weaknesses of this model since the number of topics or the approach to estimate it is rarely known, particularly for unfamiliar datasets or very large datasets.^[70–72]

2.4. Top2Vec

Top2Vec is a recently described topic modeling technique that was proposed by Dima Angelov in 2020.^[73] Unlike LDA, which uses BoW to represent documents and hence ignores word semantics, the Top2Vec method uses word embeddings. Word embedding means the vectorization of text data occurs such that the words, sentences, or documents with similar meanings are expected to be closer in spatial proximity.^[39] For example, the words such as “king” and “queen” should be closer than the words such as “king” and “apple.” The key assumption of the Top2Vec method is that semantically similar documents are indicators of an underlying topic. This technique creates joint document and word vector representations using pretrained word embeddings that capture the semantic meaning of words. As a result, the distance between these vectors represents the semantic similarity between them. Documents that are clustered together fall under the same topic.^[74] In this way, the number of prominent topics is equivalent to the number of dense document vectors. Usually, the vector space tends to be sparse, so Top2Vec first performs a dimensionality reduction on the document vectors in order to get clusters that are dense enough. Finally, the

Top2Vec model uses a clustering technique to find these dense areas and calculate the subsequent topic vectors.^[75,76]

2.4.1. Algorithm

Figure 6 shows the working mechanism of the Top2Vec algorithm. At a high level, this algorithm undertakes these steps in order to discover topics within a list of documents:

Generate Embedding Vectors for Words and Documents: The first step involves generating a joint document and word vectors using Doc2Vec. Apart from Doc2Vec, Top2Vec supports a variety of embedding strategies, including universal-sentence-encoder, distiluse-base-multilingual-cased, and BERT Sentence Transformers. The idea behind this step is to place documents such that they are close to other similar documents. The creation of embedding vectors for each of the documents allows us to handle each document as a point in the multi-dimensional space. The Top2Vec generates jointly embedded word vectors that enable us to later determine the topic keywords.

Perform Dimensionality Reduction on Vectors Using UMAP: In the next step, the model uses uniform manifold approximation and projection (UMAP) to generate lower-dimensional embedding vectors associated with each document. Since document vectors are usually very sparse in higher dimensional space, this introduces the curse of dimensionality. Each point in step 2 of Figure 6 represents a document vector.

Cluster the Vectors Using a Clustering Algorithm: A HDBSCAN is used in the third step to determine the dense area of documents. The use of HDBSCAN in topic modeling can help in further topic reduction in cases where larger topics consist of several subtopics. In Figure 6, the colored regions represent the areas with high document density, while red dots indicate outliers that do not belong to any particular cluster.

Assign Topics to Each Cluster: Finally, once the clusters are identified, each cluster of documents can be treated as a separate topic. A centroid of document vectors is determined for every cluster or dense area, which represents the topic vector. The topic is labeled by a set of keywords, which can be computed as the n -closest words to the centroid vector. In Figure 6, the purple points represent document vectors that belong to the densely populated area, and the topic vector is determined from these points (the red points are not included in the calculation of the topic vector).

2.4.2. Strengths and Weaknesses of Top2Vec

Top2Vec provides a few advantages in comparison to traditional topic modeling techniques. Its improved performance is largely attributed to its departure from convention BoW representations, which fail to capture semantic meaning and contextual relationships between words. Instead, Top2Vec leverages pretrained word and document embeddings to model semantic similarity (e.g., recognizing the relationship between “India” and “Indian,” which BoW-based models would treat as unrelated). Furthermore, by utilizing sentence-level embeddings, Top2Vec partially preserves word order, enhancing the interpretability and coherence of the resulting topics. The model also requires minimal preprocessing; there is no need for stop word removal,

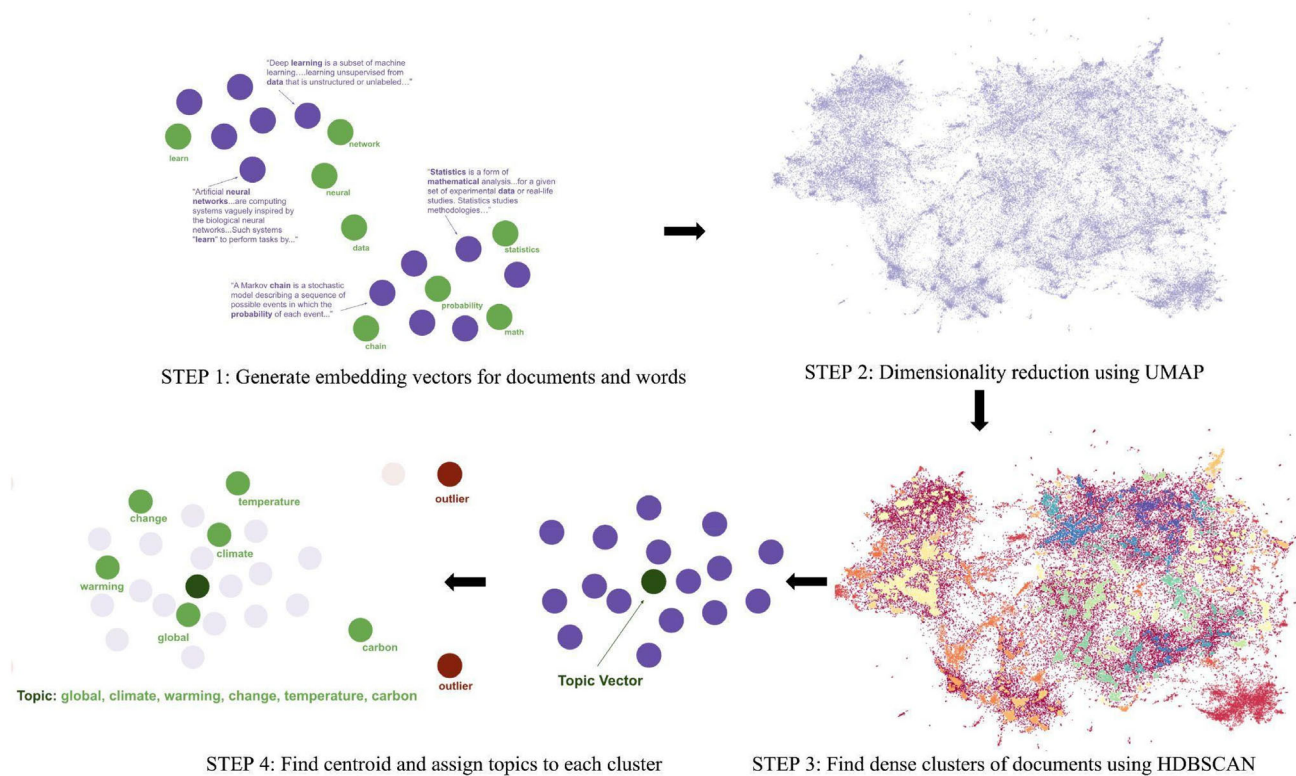


Figure 6. Algorithmic steps of Top2Vec model.

stemming, or lemmatization. Importantly, it operates without prior specification of the number of topics, enabling unsupervised topic discovery with minimal human intervention, aside from hyperparameter tuning during training. Unlike models such as LDA, Top2Vec is also well-suited to short text, making it flexible across a variety of data types. However, the absence of preprocessing introduces some drawbacks. For instance, the model may rank function words such as verbs and conjunctions among the top keywords for a topic, reducing interpretability. Additionally, Top2Vec tends to perform poorly on small datasets due to its reliance on high-quality embedding spaces, which require sufficient training data to be effective.^[39,74,77]

2.5. BERTopic

BERTopic is one of the latest techniques of topic modeling, which was introduced by Grootendorst in 2022.^[78] BERTopic operates in a similar manner to Top2Vec in terms of its mechanism and algorithmic structure; the key difference lies in the method used to select topic words. Top2Vec uses the closest word vector for topic representation. In contrast, BERTopic uses a class-based term frequency-inverse document frequency (c-TF-IDF) algorithm in order to generate coherent topic representations.^[79] More specifically, BERTopic has three steps for generating topics. First, it leverages a pretrained language model to convert documents to their embeddings. In the next step, clustering is undertaken to determine the clusters of semantically similar documents in which each cluster represents a topic.^[40] In order to improve the clustering process,

the dimensionality associated with these document embeddings is reduced. Finally, a class-based version of TF-IDF is utilized to extract the topic representation from the dense document clusters. BERTopic has many features to offer when creating the model. For example, it has the capability to support more than 50 languages by utilizing embedding models from the sentence transformers library. It creates an interactive map containing intertopic distances, which allows for individual topic examination. Similar to Top2Vec, BERTopic also has a provision for hierarchical topic reduction. Another key feature of BERTopic is that the documents that cannot be assigned to any cluster are treated as outliers.^[80,81]

2.5.1. Algorithm

Figure 7 shows the workings of the BERTopic algorithm, which consists of three major components, as discussed below.

Embedding of Documents: The documents are embedded so that they can be represented in vector space for semantic comparison. It helps to place semantically similar objects close to each other. BERTopic supports several libraries to perform the embedding of documents, including sentence transformers, flair, and gensim. Out of these, the sentence transformers library offers several high-performing sentence embedding models. The default sentence transformer model for embedding in BERTopic is paraphrase-MiniLM-L6-v2.

Dimensionality Reduction and Clustering of Documents: After building our embeddings, BERTopic compresses them into a lower-dimensional space using UMAP. Further, the HDBSCAN

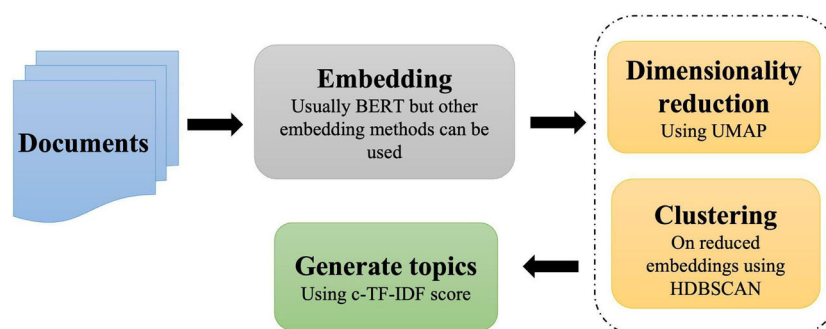


Figure 7. Algorithmic steps of BERTopic model.

algorithm is used to perform clustering in order to identify the dense clusters of semantically similar documents.

Topic Representation from Clusters: Finally, BERTopic extracts topics by assigning a topic name to each cluster by utilizing a modified version of TF-IDF that is called c-TF-IDF. TF-IDF helps in identifying or comparing the most relevant words in a given document. However, to calculate the c-TF-IDF, all of the documents within a given cluster are combined in order to create a single corpus. In this way, the c-TF-IDF score helps with the identification of the most relevant words in a cluster instead of individual documents. The higher the c-TF-IDF value of the word within the cluster, the more representative it is of the topic.

2.5.2. Strengths and Weaknesses of BERTopic

BERTopic offers several notable advantages over other topic modeling approaches. One key strength is its support for a wide range of embedding models, making it significantly more versatile than models such as Top2Vec, which offer limited embedding options. This broad compatibility enables BERTopic to perform multilingual topic modeling across more than 50 languages, greatly enhancing its applicability in diverse linguistic contexts. Additionally, BERTopic integrates rich, interactive visualizations that facilitate intuitive exploration and interpretation of topics and their associated documents. Its use of self-supervised learning techniques allows it to capture subtle linguistic patterns without requiring labeled training data. Like Top2Vec, BERTopic operates without the requirement for extensive preprocessing, such as stop word removal or stemming, and does not require prior specification of the number of topics, enabling flexible and efficient unsupervised topic discovery.^[80,82] BERTopic is a powerful framework for topic modeling, but it presents several limitations. First, it designates documents that cannot be confidently assigned to any cluster as outliers. While this can enhance topic coherence, an excessive number of outliers may lead to substantial information loss. Second, BERTopic performs hard clustering, associating each document with a single topic, which precludes the generation of topic distributions at the document level, an important feature in probabilistic models such as LDA. Third, the salient keywords within a given topic often exhibit high semantic similarity, potentially resulting in redundancy and reduced interpretability of the extracted topics.^[83,84] Further, we have summarized the characteristics of above-discussed different topic modeling techniques,

such as applicability and effectiveness, data source, dataset size, document length, level of noise, data preprocessing techniques, and vocabulary size in Table 1.

3. Topic Modeling in Different Applications

3.1. Healthcare and Bioinformatics

Miles et al.^[1] described a model that combines a reduced vector embedding representation of a pretrained language model (sBERT, or sentence-BERT is a modification of pretrained BERT model, which is used to generate sentence embeddings instead of embeddings for each word) and particle swarm optimization (PSO, a computational algorithm to find the optimal solution for a problem using population of candidate solutions) to perform the topic modeling. They applied the model to two datasets consisting of r/Cancer posts that were obtained from the Reddit archive and another dataset consisting of posts submitted to 20 newsgroups. They compared the obtained results with state-of-the-art generative models and observed that the proposed modeling technique can develop interpretable topics and has shown comparable performance to the generative models embedded topic model (ETM) and neural variational document model (NVDM). They also mentioned that sequence embedding is more suitable for topic modeling as compared to word-level embedding.

Alexander et al.^[85] utilized topic modeling as well as sentiment analysis in order to analyze healthcare data. They used patient feedback data from the Department of Health and Social Care in England that was provided by National Health Service Digital. In order to categorize each review into computer-generated topics, they applied the LDA topic modeling technique to perform the patient response investigation in three iterations. Initially, a sentiment analysis of the patient response was performed followed by a topic modeling-based analysis and in the third iteration, the topics identified, and sentiment scores were studied to associate them with each other and classify the corpus into bad and good corpora, as shown in Figure 8. Further, a well-organized channel was merged with a web application for automatic scrutiny and distribution of large volumes of information. This work represented a substantial stage to generate and visualize actionable, valuable, and exceptional data from patient reviews.

Table 1. Summary of characteristics of different topic modeling techniques.

Model	Applicability & effectiveness	Data source	Dataset size	Document length	Noise level	Preprocessing techniques	Vocabulary size	Advantages	Disadvantages
LSI	Text retrieval, topic modeling, clustering	Text documents	Large	Short to medium	Low to moderate	TF-IDF, stop-word removal	Large	Handles synonymy & polysemy, fast, dimensionality reduction	Ignores word order, low interpretability, linearity limits
pLSI	Topic modeling	Text documents	Medium	Short to medium	Low	TF-IDF, stop-word removal	Medium	Interpretable, better than LSA, probabilistic model	Not generative, poor on new docs, scalability issues
LDA	Topic modeling, clustering	Text documents	Large	medium to long	Low to moderate	TF-IDF, stop-word removal	Large	Scalable, good on sparse/unseen data, generative	Fixed topics, poor on short docs, lacks word order
Top2Vec	Topic modeling, document retrieval	Text documents	Medium to large	Short to long	Low to moderate	TF-IDF, normalization	Medium to large	Semantic & contextual modeling, no preprocessing, auto topic count	Needs large data, noisy keywords, no preprocessing
BERTopic	Topic modeling, clustering	Text documents	Medium to large	Short to long	Low to moderate	Tokenization, stop-word removal	Large	Multilingual, semantic clustering, interactive visualization	High outliers, one-topic limit, keyword redundancy

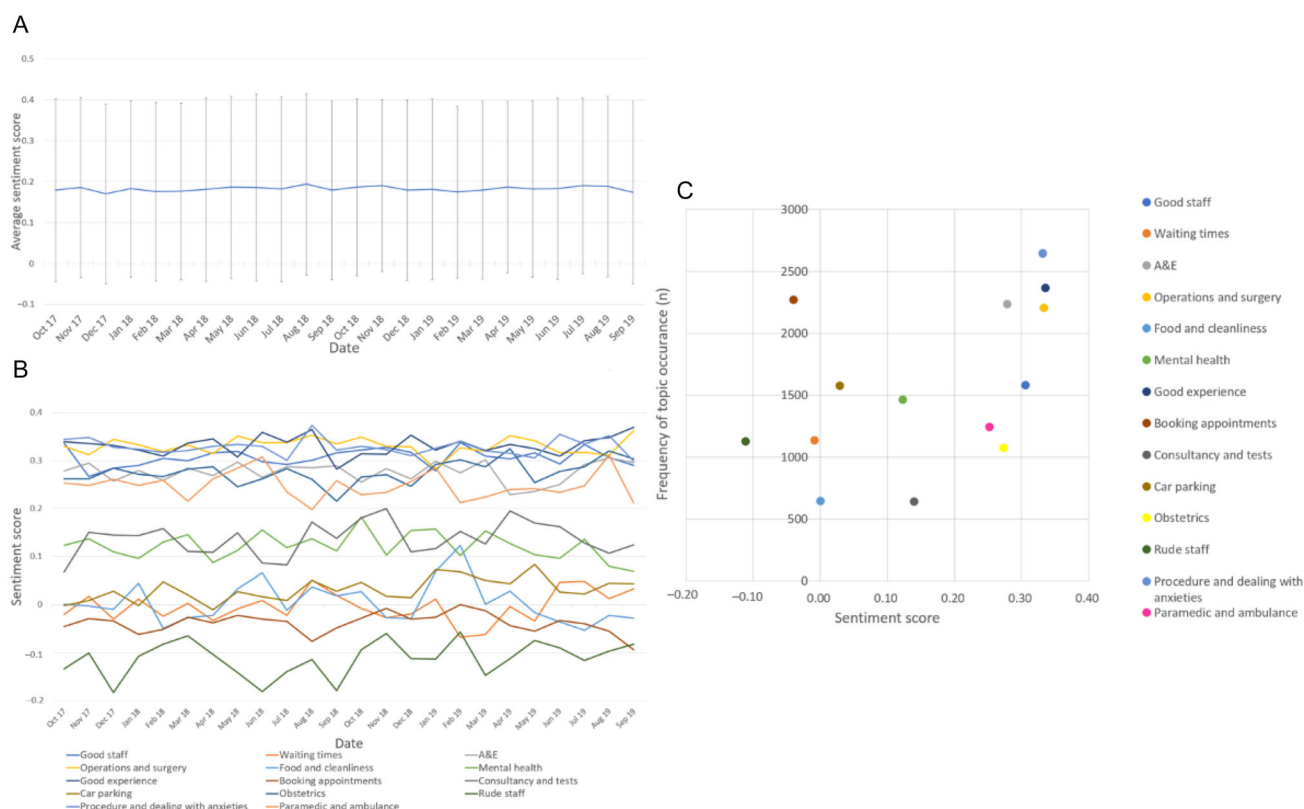


Figure 8. Illustration of relationship among A) sentiment score over time, B) sentiment score and topic frequency, and C) topic frequency and sentiment. Reproduced with permission from Alexander G., Bahja M., & Butt G. F. (2022), *JMIR Medical Informatics* 10, 4 (2022): e29385. Copyright 2022, Author(s), licensed under a Creative Commons Attribution (CC BY) License.^[85]

Rashid et al.^[86] described a novel multiple kernel fuzzy topic modeling technique (MKFTM) in the biomedical field. The model integrates the probabilistic inverse document frequency as well as the multiple kernel fuzzy c-means clustering algorithm. The probabilistic inverse document frequency method

calculates global term weights, while the MKFTM utilizes the BoW approach to estimate the frequencies of both local and global terms. The utilization of principal component analysis helps to mitigate the negative effects of higher-order term weights.

The proposed model was examined using six biomedical datasets, including MuchMore Springer, Ohsumed, Genia, Biotext, Twitter, and WSJ, and was found to outperform other models considered (e.g., LDA, LSA, fuzzy latent semantic analysis (FLSA), and Fuzzy K-means Topic Modeling (FKTM)). They noticed that MKFTM approach is effective to resolve the redundancy and sparsity issues in biomedical text documents. It also determines semantically pertinent topics with high precision and offers good outcomes for clustering and classification in biomedical documents.

Karami et al.^[87] proposed an advanced version of LSA, namely FLSA, which uses a fuzzy perspective. They applied the proposed model on five health-related datasets, namely MuchMore Springer Bilingual Corpus (M-Dataset), nursing notes (N-Dataset), Ohsumed Collection (O-Dataset), health news from Twitter (T-Dataset), and synthesis text documents in order to track the negative effect associated with redundancy in the documents (R-Dataset). They identified the underlying topics associated with the documents in the given datasets. The authors concluded that their proposed approach performed better than LDA and estimated the optimum number of topics.

Dawson and Kendzioriski^[88] created a survival supervised LDA (survLDA) framework for characterizing cancer subtypes and classifying patients into these subtypes. The dataset used in the study was obtained from the ovarian cancer cohort that was taken from The Cancer Genome Atlas (TCGA) for 448 patients. The researchers applied LDA as well as survLDA and specified the number of topics as seven for all analyses. They extended the framework to permit supervision by a time-to-event response. The authors found that the proposed approach offers a means of efficiently identifying collections containing clinical features and genomic features that are found together in patient subgroups as well as characterizing each patient based on these features.

Zhao et al.^[89] applied topic modeling in the field of biological research or medical research. The authors utilized three significant datasets, which represent diverse types of large biological datasets or medical datasets, to demonstrate the use of topic modeling in cluster analysis. These datasets are the Salmonella PFGE genotyping data obtained from the CDC public dataset, the breast cancer microarray dataset, and the lung cancer microarray dataset obtained from the Gene Expression Omnibus. They carried out LDA topic modeling, followed by three distinct cluster analysis techniques, on the topic model-generated data matrices from the three datasets. The authors concluded that using topic modeling for cluster analysis with large datasets can significantly enhance the precision and effectiveness of subgroup identification.

Paul and Dredze^[90] described a statistical topic modeling framework to study public health information through millions of health-associated tweets. The authors obtained two Twitter datasets, with the first dataset consisting of greater than two billion tweets ranging from May 2009 to October 2010; the second dataset included a daily average of four million tweets from August 2011 to February 2013. They finally experimented on the tweets that were selected based on 269 health keywords. Topic modeling was performed using two different models, namely LDA and ailment topic aspect model (ATAM). The authors found that utilizing topic models aid in discovering various health issues that have a significant correlation with actual

surveillance and survey data, in spite of using negligible human management and no historic information to train the model.

Mohammed and Noorullah^[91] described an extended version of the LDA model, namely the ATAM and multiaspects topic model (MATM), to obtain effective healthcare recommendations using Twitter data that considered health-related topics. The authors performed experiments on benchmarked health data provided by TREC 2015 and TREC 2014 to recommend healthcare solutions with multiple aspects. The proposed MATM model provided healthcare recommendations with fewer side effects based on highly distributed symptoms of a particular disease and showed better performance than ATAM. They also derived the less influential treatments for the supplementary relationship of development between health recommendations and diseases, along with healthcare data clustering.

Baird et al.^[92] used machine learning (ML) and topic modeling in order to understand the principal topics of consumer discussion on Twitter that were associated with telehealth for substance abuse or mental health during the pandemic and prepandemic. To perform this study, the authors utilized BERTopic and Twitter data collected over the period between January 2014 and June 2021 through the use of keywords related to telehealth such as telemedicine, telehealth, and virtual care followed by describing the final corpus, optimizing BERTopic in Python, and refining thematic classification and topic model results. The authors observed that BERTopic is more effective than ML in extracting the relevant topics. Also, it has been observed that the number of tweets associated with telehealth for mental health and/or substance abuse rose by fourfold during the pandemic.

Meaney et al.^[93] proposed a framework to compare different types of topic modeling techniques, namely LDA, NMF, structural topic model (STM), and BERTopic using primary care clinical text data. The authors utilized clinical notes taken from electronic medical records associated with primary care in Ontario, Canada, covering the period from January 01, 2011, to December 31, 2015. They generated heat map illustration of the multivariate time series structure using different temporal topic models, an example of which is shown in **Figure 9**, created with BERTopic. The study showed that topic models could provide a method for utilizing unstructured clinical text data to perform characterization and surveillance of primary care practices and primary care systems. It was observed that NMF, LDA, STM, and BERTopic need unique hyperparameters tuning and have separate computational necessities, such as computational hardware and data structures. Nevertheless, all models reliably assessed a latent variable representation of the primary care text data, which implicitly categorized primary care patients and their communications with the primary health care system.

Recently, Ashiquil et al. applied mutational signature analysis in cancer genomics using an automated framework, SigProfilerExtractor.^[94] The tool performed de novo extraction of mutational signatures without reliance on predefined reference signatures, directly from somatic mutation data. To achieve this, the study employed NMF to decompose the high-dimensional mutation count matrix into two lower-rank non-negative matrices: one representing the mutational signatures and the other capturing their respective contributions across samples. Through extensive benchmarking against 13 existing tools, SigProfilerExtractor

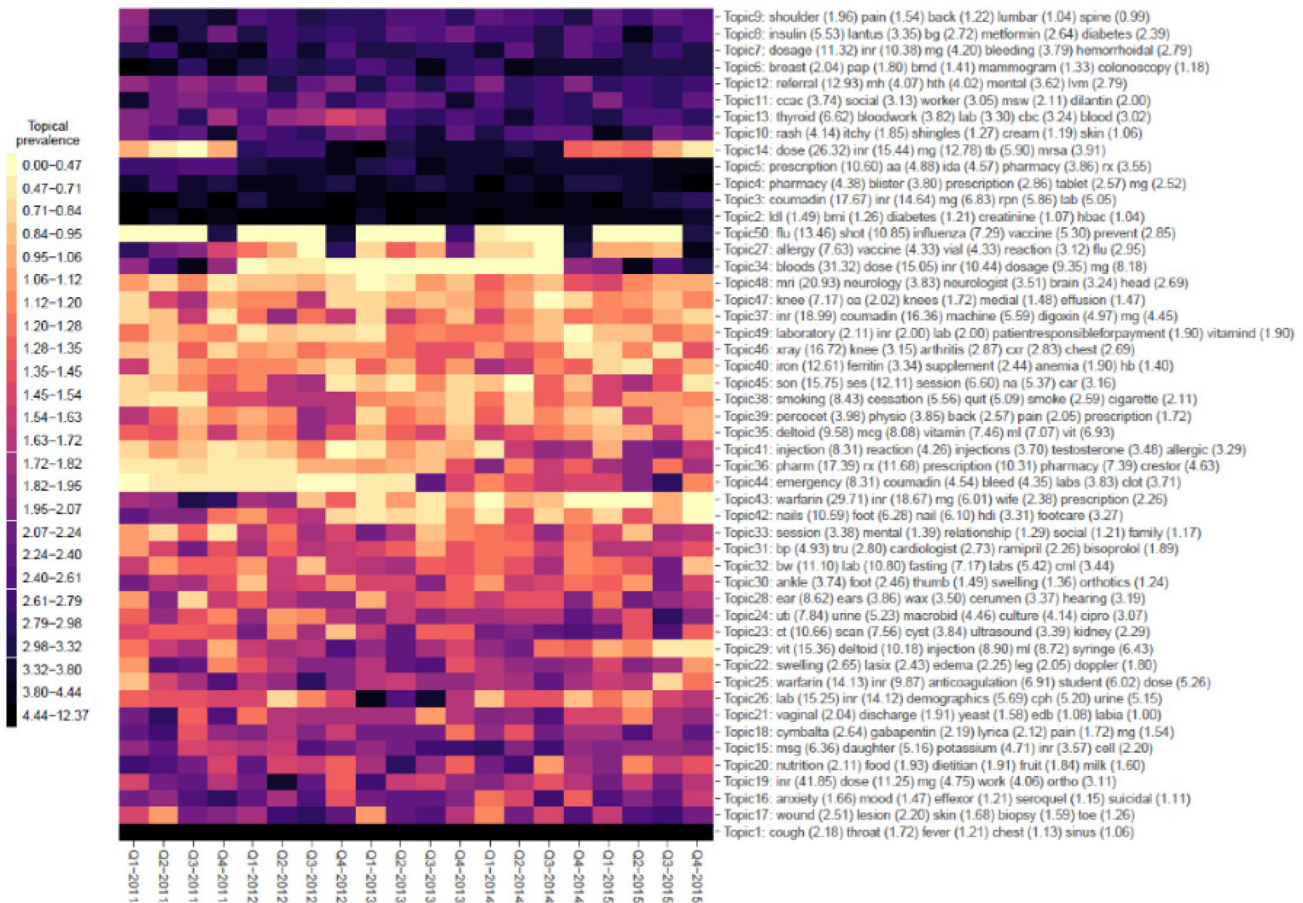


Figure 9. Illustration of heat map diagram of the multivariate time series structure associated with the BERTopic temporal topic model. Reproduced with permission from Meaney, C., Escobar, M., Stukel, T. A., Austin, P. C., & Jaakkimainen, L. (2022). *JMIR Medical Informatics* 10, 12 (2022): e40102. Copyright 2022, Author(s), licensed under a Creative Commons Attribution (CC BY) License.^[93]

demonstrated superior performance and successfully identified four previously unreported mutational signatures, highlighting its robustness and effectiveness in uncovering novel mutational processes.

Similarly, Pelizzola et al. evaluated the use of topic modeling to infer mutational signatures from cancer genomic data, where each sample is represented as a count matrix of distinct mutation types.^[95] Departing from the conventional application of NMF, the authors highlighted key limitations in assuming continuous and normally distributed data, noting that mutational counts are discrete, exhibit overdispersion, and deviate from Gaussian assumptions. To address these issues, they introduced a novel framework, Negative Binomial NMF, which models the count data more accurately by incorporating the negative binomial distribution. The study places particular emphasis on determining the optimal number of mutational signatures, employing extensive simulation experiments and robust model selection criteria to validate the approach.

In another study, Donker et al. emphasized the inherent complexity of healthcare data, encompassing heterogeneous modalities such as gene expression profiles, somatic mutation patterns, clinical phenotypes, and risk factors.^[96] The authors argue that

the intricate interactions among latent topics in such high-dimensional, multisource data are not effectively captured by traditional models like NMF. Instead, they advocate for Bayesian approaches, specifically MBNs, which extend the capabilities of LDA. MBNs model count data using multinomial distributions while introducing a hierarchical structure of latent variables, allowing for more expressive representations and improved inference in complex biomedical contexts.

Rosenthal et al. demonstrated a comparative study to evaluate their tool, DeconstructSigs, against existing frameworks for analyzing somatic mutations shaping the cancer genome.^[97] The model was designed to deconvolute mutational processes and identify mutagenic exposures contributing to carcinogenesis, with the goal of reconstructing the complete mutational landscape at the individual tumor level. In contrast to previous approaches that typically rely on large cohort analyses, DeconstructSigs emphasizes sample-specific resolution, enabling a more granular characterization of mutational signatures in single tumors. The method employs a multilinear regression framework, where the mutational profile of a tumor is described as a linear combination of predefined reference signatures, with regression coefficients estimating their relative contributions. Although not a traditional topic modeling

technique, DeconstructSigs shares conceptual similarities in its goal of uncovering latent structure within high-dimensional mutation data.

Building upon prior research, Degasperis et al. proposed a comprehensive framework for mutational signature analysis, emphasizing inter-organ and inter-tissue heterogeneity as well as the dependency of mutational processes on specific driver mutations.^[98] Their study revealed that while some mutational signatures are conserved across multiple tissues, others exhibit strong tissue or organ specificity. Furthermore, the analysis uncovered significant associations between certain driver mutations and distinct mutational signatures, suggesting mechanistic links between genetic alterations and mutational processes. The framework incorporates NMF as the core extraction method, with comparative evaluations of various optimization strategies and clustering algorithms to enhance the robustness and interpretability of signature identification across diverse biological contexts.

McCoy et al. implemented LDA, a generative probabilistic model, on electronic health record (EHR) data from 10 845 individuals within a biobank context.^[99] LDA was employed to uncover 50 latent topics that represent clusters of co-occurring diagnostic codes. The weights of these topics were then utilized as quantitative phenotypes in genome-wide association studies (GWAS), resulting in the discovery of several known and new genetic loci linked to complex diseases. The research showed that phenotyping based on topics can enhance the effectiveness of GWAS by more accurately capturing the underlying structure of comorbidities compared to traditional methods that focus on single diagnoses.

Further, Jiang et al. presented age-dependent topic modeling (ATM), which builds upon LDA by integrating age as a variable to capture the temporal patterns of disease co-occurrence.^[100] When applied to longitudinal EHR data from 282 957 participants in the UK Biobank, ATM revealed ten disease topics that reflect comorbidity patterns dependent on age. This model enabled the identification of disease subtypes based on the assigned topics, uncovering 52 diseases with varied comorbidity characteristics. Following genetic investigations, 18 disease subtypes were identified, each with notably different polygenic risk scores (PRS) and particular genetic variants that displayed effects dependent on the subtype, underscoring the genetic diversity within common diseases.

Zhang et al. introduced treeLFA (latent factor allocation guided by a tree-structured prior), a framework for topic modeling that incorporates hierarchical connections from the ICD-10 coding system into its generative process.^[101] By leveraging data from the UK Biobank, treeLFA was able to identify clusters of multimorbidity, including a “healthy” category, through the modeling of disease co-occurrence patterns. The topic weights obtained from treeLFA served as traits in GWAS, revealing new genetic associations that partially coincided with those discovered in single-disease GWAS. The research showed that the integration of structured priors based on medical ontologies could improve the interpretability and predictive capabilities of topic models in understanding complex disease interrelationships.

Conclusively, this section discusses research efforts aimed at advancing classical topic modeling techniques through the integration of sentence-level embeddings for improved contextual understanding and the application of fuzzy logic to

mitigate issues like redundancy and sparsity. The reviewed studies emphasize practical applications in enhancing information retrieval, diagnostic precision, and biomedical research. Additionally, the role of topic modeling in active surveillance systems, augmented by the expanding body of medical knowledge, highlights its potential in preventive healthcare and personalized treatment strategies. Collectively, these advancements point toward the development of sophisticated topic modeling systems that are poised to transform the future landscape of telehealth and primary care.

3.2. Scientific Research Articles

Choi et al.^[102] utilized the LDA model to perform the topic modeling in order to identify the latent trends in research. They extracted the abstracts of 2356 documents that include journal articles, conference papers, book chapters, reviews, and working papers, which were published between 1972 and August 2015. They found that the topics uncovered in all the documents primarily center around technology; the results show that algorithms, privacy on Facebook, and online social networks have emerged as significant topics. In comparison, journal articles gave a greater level of attention to e-business and healthcare.

Chakkarwar and Tamane^[103] applied the LDA model with BoW to identify underlying themes, patterns, and the latest trends in order to consider different research trends using topic modeling. They applied the proposed model to a dataset of abstracts from 550 IEEE publications involving the field of blockchain technology to get an insight into the research literature. The topics created by the LDA model (using a generative probability distribution approach) demonstrate numerous hidden themes, connected terms to the main theme of documents, providing a prompt summary of these documents.

Anantharaman et al.^[104] utilized three different topic modeling approaches, namely LDA, LSA, and NMF, for short text or sentence classification as well as large text or document classification on openly accessible news datasets. The models were evaluated on three datasets, including PubMed, 20 k RCT BBC news dataset, as well as a subset of the Twenty Newsgroups dataset. Based on the observations, they indicated that LSA is the most effective topic modeling algorithm for short text, while LDA accomplishes better on large text. They also observed that LSA is associated with a lower level of execution time as compared to LDA. Further, they mentioned that a tradeoff between precision and running time is a vital point to be considered while determining the application-specific performance of topic modeling techniques.

Walker et al.^[105] used topic modeling to understand the research-practice gap in the field of public administration. The authors used LDA, a well-known topic modeling technique, in order to examine the content of the abstracts and titles of Public Administration Review (PAR) and PA Times (PAT) articles. They observed that greater than 50 percent of topics were common topics to both groups, which showed shared interests. It should be noted that they also found some topics that were distinct for the two groups. The outcomes revealed a variety and not a disengagement.

Kukushkin et al.^[106] described a method of visualizing the landscape of digital-twin science with bibliometric tools, text-mining, and topic modeling with the help of LDA and BERTopic (Bidirectional Encoder Representations from Transformers). They analyzed a dataset of 8693 publications associated with the topic that were collected from the Scopus database, covering the period from January 1993 to September 2022. The authors found that the LDA model identified numerous key topics; however, their interpretation was not clear. On the other hand, the BERTopic model discovered greater than 100 topics; eight of these topics were considered the most significant. The authors concluded that research in this area remains at an early stage; however, various theories and methods are converging on a distinct field of study.

Abuzayed and Al-Khalifa^[80] applied topic modeling in order to determine abstract topics from a large collection of Arabic documents. The authors used an advanced BERT-based technique, namely BERTopic, with different pretrained Arabic language models as embeddings; they compared the results of this technique against the LDA technique and NMF technique. They used the “dataset for Arabic classification” that contained 111 728 documents that were written in modern standard Arabic (MSA), which was obtained from three Arabic language online newspapers, including Hespess, Assabah, and Akhbarona. They found the performance of BERTopic to be better than NMF and LDA.

Hendry et al.^[75] used topic modeling on customer service chat data to understand new intents in the user messages, which were not previously recognized, and reorganized existing intents in the messages. Three topic modeling techniques were utilized: Top2Vec, LDA, and BERTopic. The study was conducted on five datasets, including four English public datasets (ATIS, SemanticParsingDialog, CLINC150, and Snips) as well as one nonpublic Indonesian dataset (e-commerce). The results showed that BERTopic and Top2Vec provided enhanced evaluation scores in comparison to LDA in the customer service chat dataset. The authors concluded that their methodology can be used further to analyze chatbot message datasets.

Silveira et al.^[107] analyzed legal documents using topic modeling. They constructed a BERTopic model utilizing domain-specific embeddings from a pretrained LEGAL-BERT model. The dataset consisted of a collection of legal documents from the historic US Supreme court decisions repository (Cornell Legal Information Institute), which contains significant court decisions in the United States. The authors found that the proposed framework performed well and uncovered topics that align with the document’s topics, providing a foundation for future research.

Scarpino et al.^[108] applied LDA and a BERT transformer-based topic modeling approach to examine the Italian perspective on the COVID-19 pandemic. They used 187 narratives that were obtained from two Italian projects, called Sindrome post COVID-19 and R-esistere, which were divided into PASC and non-PASC testimonies. The study found that the BERT-based approach showed higher effectiveness than the LDA approach; it was able to group 97.26% of the analyzed documents into the same cluster, resulting in an overall accuracy of 91.97%.

Atzeni et al.^[109] described a topic modeling framework to systematically review the interaction/integration between Wi-Fi and ML. To perform this study, authors extracted abstracts of papers

from Web of Science, Scopus, and IEEE Xplore databases and used a topic modeling technique, called BERTopic, to examine the document corpus. Based on the obtained topics, they determined that the combination of ML and Wi-Fi data has influenced many fields.

Zou et al.^[110] proposed a PWF model that combines a typical probabilistic topic model, known as PLSI, with website fingerprinting (WFP) in order to monitor online criminal activities. The proposed model creates a topic probability vector to obtain new types of features for each traffic instance. Based on these vectors, a KNN classifier was utilized in order to undertake a website fingerprinting attack.

Harashima and Kurohashi^[111] proposed a framework using PLSI to perform topic modeling and summarize the results. To perform the experiments, the authors studied IT workers, who were asked to complete a questionnaire. These workers were asked to use this system to understand four aspects associated with the proposed approach. The validity of the degree of redundancy, number of topics, precision of clustering, and effectiveness of the method in presenting the summary were considered. The observations demonstrated that this method efficiently classified search results into topics.

This section highlights the core objective of topic modeling, leveraging diverse textual data sources to uncover latent sub-topics, and detect emerging trends. Comparative studies on the performance of classical modeling techniques lay a strong foundation for future innovations and enable more refined, domain-specific applications. Furthermore, the exploration of novel use cases including chatbots, multilingual topic models, legal assistance systems, and online criminal activity surveillance, underscores the broad and growing potential of topic modeling across a wide range of domains.

3.3. Social Media Platforms

Alshsalan et al.^[112] described a deep learning and topic modeling-based approach to detect hate speech within tweets that were related to COVID-19. The authors utilized a pretrained convolutional neural network (CNN) model to analyze the tweets for hate speech; they assigned a score between zero and one to each tweet. In this approach, the score of one was assigned to the text with the most hateful attributes. Further, they applied a NMF model to understand the issues and topics that were considered in hate tweets. To conduct this study, the authors used the ArCOV-19 dataset, a collection of Arabic tweets that were related to COVID-19.

Liu and Jansson^[113] performed the topic modeling analysis of Instagram data, a social media platform. They applied the LDA topic modeling technique on the English data to understand the overall topic presence and prevalence in the data. The dataset employed in this work was collected over a period of three months from public Instagram comments and posts that were obtained from the greater Helsinki region. The authors concluded that topic modeling helped them identify the most relevant latent topics through Instagram posts, mainly concerning trips, travel, and vacations in Finland.

Albalawi et al.^[114] compared different available topic modeling algorithms, including LDA, LSA, random projection (RP), NMF,

and principal component analysis (PCA). They used two publicly available datasets, namely 20NewsGroups and short conversation data that were obtained from the Facebook social network site, in order to examine the functionality of the proposed models. They concluded that LDA and NMF provided the most meaningful and valuable outputs over diverse ranges.

Zankadi et al.^[114] utilized topic modeling and NLP techniques to predict the actual interests of MOOC learners based on the text content they shared on social media. The authors applied three topic models, including LSA, LDA, and BERTopic. The study used a dataset of 120 000 tweets from 12 187 learners scraped from Twitter based on the keywords “artificial intelligence” and “computer science.” The results indicated that BERTopic outperformed LDA in terms of the relevance and the quality of the topical interests that were more representative of the course of interest of the learners.

Xu et al.^[115] applied LDA to perform topic modeling on Douban online review data of the MCU, which contains some reviews in English and some reviews in traditional Chinese characters. Through topic modeling, authors were able to identify the reasons that underlie the success of the MCU and to help Chinese movie creators understand the psychological needs associated with movie viewers.

Asghari et al.^[116] proposed an adaptive system that was designed using five layers, which combined unsupervised and supervised algorithms in order to track health trends that were on social media. The authors performed this study on a Twitter dataset of 144 922 tweets from the USA, which were collected based on the following keywords: health, healthcare, homecare, digital health, and doctors. They applied the LDA topic modeling technique in order to identify the latent topics in the tweets. This study helped to detect the most trending topics related to healthcare in each state of the USA.

Ventura et al.^[117] compared eight topic modeling and clustering algorithms for health-related tweets and health-related emails. These algorithms included LSI, online LDA, online Twitter LDA, LDA, LDA with Gibbs sampling (GibbsLDA), Gibbs sampling for Dirichlet multinomial mixture (GSDMM), and Biterm model (BTM) using two different embeddings: TF-IDF and Doc2Vec. They concluded that online Twitter LDA and GSDMM provided the highest performance for deriving the clusters and topics.

Egger and Yu^[118] compared four of the topic modeling techniques, namely NMF, LDA, Top2Vec, and BERTopic. The comparison was conducted using a dataset of 50 000 tweets related to travel and the COVID-19 pandemic, which was obtained through the data extraction software tool Phantombuster using keywords such as “covid travel,” “covid,” and “travel” for the month of November 2021. The authors found that BERTopic showed greater potential in producing topics from short and unstructured text in comparison to the other topic modeling methods that were considered.

In summary, social media provides a rich source of data for analyzing a wide range of topical interests. This section highlights several key applications, including hate speech classification, extracting actionable insights from reviews and comments, and survey analysis. In addition, the research presents a thorough comparative evaluation of prominent topic modeling approaches, with a particular focus on their strengths and effectiveness in generating high-quality, relevant, and diverse topics.

3.4. News Articles

Ray et al.^[119] applied different topic modeling techniques, NMF, LSI, and LDA algorithms, to perform topic modeling in the Hindi language. They used a dataset available on GitHub, which contains news articles from two Hindi newspapers with the titles Amar Ujala and Navbharat Times. The authors concluded that processing documents in the Hindi language for information retrieval is challenging due to the complexity and certain inherent characteristics of the Hindi language. Moreover, less software and tools are available to support NLP tasks in Hindi, making the task even more challenging. The application of the above-mentioned topic modeling techniques helped to discover the topics of the newspaper articles.

Zhou et al.^[120] proposed a graph topic model (GTM), which is a GNN-based neural topic model in which a corpus is represented as a document relationship graph. They applied the proposed model to three different data sets, including 20NewsGroups, Grolier, and NYTimes documents. Moreover, they compared the performance provided by the proposed model with LDA NVDM, ProLDA, GraphBTM, ATM, and W-LDA to understand the effectiveness of the proposed approach.

Mazarura and de Waal^[121] proposed the GSDMM to perform topic modeling on short text. To perform the experiments, they used three different text corpora, including Finweek news articles, Crowd Flower Open Data Library, and tweets that involved a GOP debate that occurred in Ohio. They compared the performance of GSDMM and LDA; they concluded that LDA in general outperformed GSDMM for long text. For short text, LDA outperformed GSDMM for one case; GSDMM outperformed LDA for the other case.

Chen et al.^[122] used LDA and NMF and their variants, such as BTM, LFLDA, GPUDMM, GKLD, and KGNMF, in order to undertake the topic modeling on short texts. They performed the experiments on five different datasets, namely snippet, XinlangNews6, News, TMNtitles, and StackOverflow7. They observed that NMF-based approaches performed better than LDA-based approaches in topic mining with short texts. Moreover, their proposed KGNMF, which is a knowledge-guided model, performed comparatively better than the other models.

Liu et al.^[123] evaluated the role of the media and the patterns of health communications in the early stage of the COVID-19 pandemic in China using topic modeling techniques. The authors applied the LDA topic modeling technique on a corpus of news articles regarding COVID-19 that was released by the WiseSearch database between January 1, 2020, and February 20, 2020. The obtained results are categorized as one of nine main primary topics; the top three topics are as follows: medical research and treatment, local or global influences of a social and economic nature, and prevention and control procedures. Further, they demonstrated (Figure 10) the top 30 most pertinent terms of topic 1 using the topic modeling approach, having the highest proportion of all topics. Thus, the authors concluded that using topic modeling of news articles may be helpful in generating important information for early health communication.

Conclusively, the research presented in this section offers a comprehensive evaluation of topic modeling performance across a diverse array of datasets, including news articles,

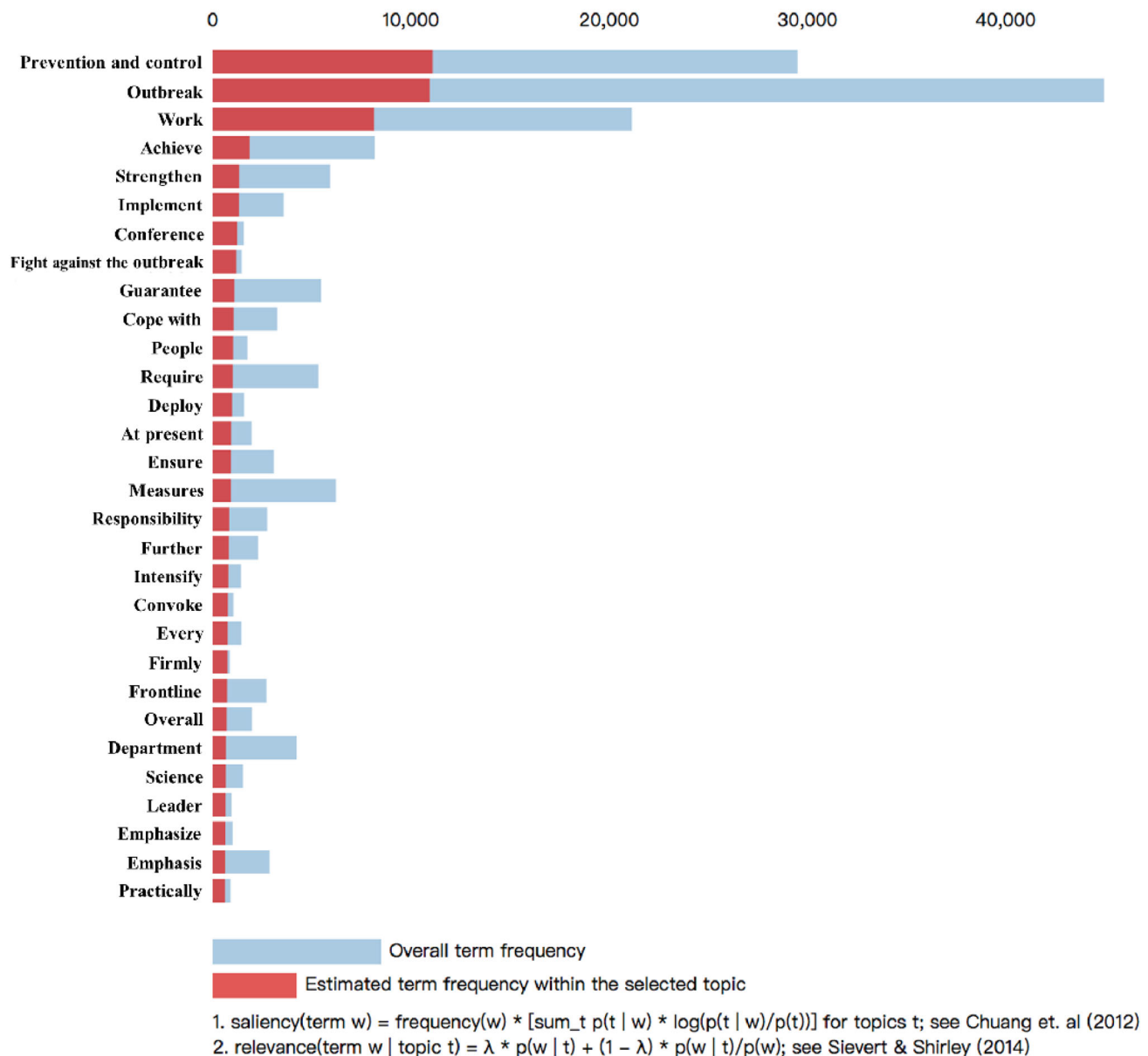


Figure 10. Illustration of Top 30 most relevant terms for Topic 1 generated using the topic modeling approach. Reproduced with permission from Liu, Q., Zheng, Z., Zheng, J., Chen, Q., Liu, G., Chen, S., Chu, B., Zhu, H., Akinwunmi, Ba., Huang, J., Zhang, C. J. P., & Ming, W. K. (2020), *Journal of medical Internet research* 22, 4 (2020): e19118. Copyright 2020, Author(s), licensed under a Creative Commons Attribution (CC BY) License.^[123]

encyclopedia entries, topic summaries, multilingual news corpora, forum-based Q&A, and article titles used for benchmarking. Furthermore, the innovative integration of deep learning techniques, advanced data structuring, and the incorporation of external knowledge sources effectively addresses several limitations inherent in classical topic modeling approaches.

3.5. Legal Domain Documents

O'Neill et al.^[124] performed an analysis that utilized topic modeling on legislative data of British legislation to summarize and visualize salient legal topics. The authors used a dataset of a

corpus containing 41 518 documents of legislative texts of the UK for a duration of 16 years from 2000 to 2016. To perform the topic modeling, the authors used LDA, pLSA, and NMF models. Based on the obtained results, the authors determined that Standard LDA showed the best results when compared with other types of state-of-the-art topic modeling techniques.

Viksna et al.^[125] demonstrated the application of different topic analysis algorithms to the legal document corpus to discover expressive and meaningful legal topics. They applied the algorithms to a corpus of legal acts documents from a website of legal acts from the Republic of Latvia. They applied hierarchical Dirichlet process (HDP), LSA, and LDA for topic modeling.

The authors indicated that in comparison to the LSI and the HDP, the LDA algorithm produced a topic model, which was shown to perform better than the alternatives.

Araujo and Campos^[126] proposed a topic modeling framework for modeling a corpus consisting of extraordinary appeals that were received by the Supreme Court of Brazil. They applied LDA on a corpus containing 45 532 lawsuits that was manually annotated by the experts of the court with multi-label and multi-class topic labels. The LDA models were initially trained with 10 and 30 topics to analyze the most relevant keywords of each topic. Further, the model was trained with 30, 100, 300, and 1,000 topics in order to generate feature vectors associated with each appeal, which were further utilized to train a lawsuit topic classifier. Based on the observed results, the authors concluded that the model containing 300 topics achieved the best classification results.

In the study by Chen et al.,^[127] the authors compared various topic modeling techniques, including NMF, LDA, PCA, and k-competitive autoencoder (KATE), to uncover latent topics within a large collection of SEC writing text documents that were produced by US public banks. The data analyzed consisted of 8-K and 10-K writings of 578 of the bank holding companies, covering the period from 2005 to 2016. Through their analysis, the authors discovered meaningful topics in the documents and determined that topic modeling can play a crucial role in terms of financial decision making and risk management.

Salun et al.^[128] utilized topic modeling to extract the latent topics that underlie large corpora of judicial texts. They developed topic models to identify the reasons a tenant prosecutes their landlord in court. The authors applied two traditional topics and one neural topic model, such as LDA, LSI, and BERTopic on 12 102 cases spanning 2001 to 2018 obtained from the Régie du logement du Québec.

Concludingly, this section discusses research focused on the unique challenges and innovations in applying topic modeling to legal texts. Legal documents often feature dense language, complex terminology, and ambiguous topics, making interpretability and coherence critical. Many of the reviewed studies effectively address these issues by improving the models' ability to interlink and contextualize legal cases and documents. Additionally, the simplification of complex legal taxonomies emerges as a promising direction, enabling more intuitive interpretation and querying. These contributions highlight the ongoing advancements in adapting topic modeling for the intricacies of the legal domain.

4. Evaluation Metrics

Evaluation of ML models is one of the most important tasks for evaluating the potential and quality of developed models. Similarly, different types of evaluation metrics are utilized to understand the performance of topic models. The metrics examine the coverage, coherence, as well as quality of the discovered topics. Some of the most popular evaluation metrics are discussed below.^[129]

4.1. Coverage

The extent to the topics identified by topic models accurately reflect the content within the document collection is referred

to as coverage. Coverage is used as an evaluation metric for topic models. It is particularly useful when the goal involves identifying a comprehensive set of topics that cover as much of the input text as possible.^[130] A few of the most used evaluation metrics that measure coverage include topic recall, topic model accuracy, purity, and perplexity; these metrics are discussed below in detail.^[131]

4.1.1. Topic Recall

Topic recall measures how well a topic model can identify all of the relevant topics in a given dataset. It is determined considering the number of relevant topics identified by the model divided by the total number of relevant topics in the dataset. Topic recall is often utilized as an evaluation metric for the topic models. It is particularly useful when the goal is to identify as many relevant topics as possible in a dataset.^[132]

$$\text{Recall} = \frac{\text{Number of relevant topics retrieved}}{\text{Total number of relevant topics}} \quad (7)$$

4.1.2. Topic Model Accuracy

Topic model accuracy measures how well a topic model can classify new documents into the appropriate topic. It is calculated using the number of documents that are classified correctly by the model, which are divided by the total number of evaluated documents. Topic model accuracy is often used as an evaluation metric for topic models; it is particularly useful when the goal is to classify the new documents into the appropriate topic. To assess the recall and accuracy of topics, it is necessary to have ground truth topics available.^[133]

$$\text{Accuracy} = \frac{\text{Number of correctly classified documents}}{\text{Total number of evaluated documents}} \quad (8)$$

4.1.3. Purity

Purity measures the degree to which a topic is composed of a single topic in topic modeling. A topic with high purity will be focused on a single topic, while a topic with low purity may contain a mix of unrelated or tangentially related words. Purity is often utilized as an evaluation metric for topic models and is particularly useful when the goal is to identify clear and coherent topics. This approach is utilized in situations in which a complete set of ground truth topics is not readily accessible.^[134]

$$\text{Purity}(\Omega, C) = \frac{1}{N} \sum_k \max_j |\omega_k \cap c_j| \quad (9)$$

In this equation, $\Omega = \{\omega_1, \omega_2, \dots, \omega_K\}$ represents the set of clusters and $C = \{c_1, c_2, \dots, c_J\}$ represents the set of classes. We interpret ω_k as the set of documents in cluster ω_k and c_j as the set of documents in class c_j .

4.1.4. Perplexity

This metric considers the ability of the model for predicting the probability of a held-out sample of documents being assigned to a particular topic. A lower level of perplexity is associated with a better model, as it can accurately predict the likelihood associated with a given document. The perplexity score can be calculated by taking the exponent of the average cross-entropy of the model across all documents in the corpus^[119]

$$\text{Perplexity}(D_{\text{test}}) = e^{\frac{\sum \log p_{w_d}}{\sum \log N_d}} \quad (10)$$

where, w_d - words in document, N_d - number of document.

4.2. Coherence

Coverage measure is a significant criterion to assess the quality of identified topics; however, it alone is not sufficient to validate the performance of topic models. Besides coverage, researchers use various evaluation metrics to measure the coherence. Coherence measures the degree of usefulness and interpretability of the topics and the keywords in the topics generated by models.^[134] The coherence value of a topic model is high if the words in a particular topic are semantically similar and make the topic meaningful. A higher coherence score is associated with a better model. Precision, diversity, and pointwise mutual information (PMI) are the most popular evaluation metrics to measure coherence; these metrics are discussed below in detail.^[135]

4.2.1. Precision

Precision measures the proportion of words within a topic that are relevant to that topic. Precision can only be calculated if the ground truth topics and the keywords that are associated with each topic are already established. It is determined as the proportion of relevant words in a topic to the total number of words that are associated with that topic. A higher precision value indicates a better topic model with meaningful and interpretable topics.^[136]

$$\text{Precision} = \frac{\text{Number of relevant words in topic}}{\text{Total number of words in topic}} \quad (11)$$

4.2.2. Diversity

Diversity measures the degree of distinctness between different discovered topics via the topic model. A model that produces a wide variety of topics that are unique and not similar to each other is said to have high diversity.^[137] A model that generates similar topics must have an overlap between keywords of the topic. Topic diversity can be determined by calculating the percentage of distinctive words within the set of topics. Diversity is often used as an evaluation metric for use with topic models; it is particularly useful when the goal involves identifying a diverse set of topics within a dataset^[138]

$$\text{Diversity} = \frac{|\cup \text{TopWords}(k)|}{K \times N} \quad (12)$$

where K is the number of topics and N is top- N words

4.2.3. Pointwise Mutual Information (PMI)

The PMI measures the association between two words within each topic-based on their relative frequencies in relation to each other. PMI is calculated as the log of the fraction of the joint probability associated with two words occurring together to the product of their individual probabilities.^[139] A higher PMI score indicates that the two words are highly associated with each other. PMI is more popular in the topic model as it does not require any information associated with the ground truth topic, unlike precision. It is used with other evaluation metrics, such as coherence and perplexity, in order to understand the performance of a topic model^[140]

$$\text{PMI}(w_i, w_j) = \log \frac{P(w_i, w_j)}{P(w_i) \times P(w_j)} \quad (13)$$

w_i - word i , w_j - word j

4.3. Train-Test Splits in Topic Modeling

4.3.1. Reason for Different Train-Test Split for Topic Modeling

Evaluating topic models differs from traditional supervised learning due to their unsupervised nature. Unlike predictive models, topic models aim to discover coherent topics and meaningful topic distributions across documents. As a result, conventional random train-test splits may not be ideal, as they can disrupt the model's ability to uncover global topic structures.^[141]

4.3.2. Recommended Strategies for Train-Test Splits

A common and effective evaluation strategy involves holding out a subset of documents to assess the model's generalization capability. This "held-out documents" approach measures how well the model assigns topic probabilities to unseen documents. It's crucial to use evaluation metrics tailored for topic modeling, such as perplexity and topic coherence, rather than accuracy.

4.3.3. Training versus Evaluation: Estimating Topic Model Components

In practice, the model is trained on a training set to estimate parameters like the word-topic distribution (Φ) and the document-topic distribution (Θ) for training data. When evaluating, Φ is kept fixed, and Θ is inferred for the held-out set. Metrics like perplexity are then calculated to evaluate how well the learned topics apply to new data.

4.3.4. Conclusion

Train the topic model on the training set to learn the topics, then use the held-out set to assess how well those learned topics generalize to new documents. The key is to evaluate the model based

on its ability to assign probabilities to unseen data, rather than predictive accuracy in a supervised sense.

4.4. Visual Inspection

Quantitative metrics such as coverage and coherence can provide insight related to the quality of a topic model; however, human evaluations offer a more comprehensive view. They consider multiple criteria, including interpretability and relevance, and provide a domain-specific perspective.^[142] Visual inspection of the identified topics and the keywords associated with them is one of the most effective approaches for evaluating the quality of a model. Evaluation through visual inspection in topic modeling involves presenting the topics generated by the model to human evaluators for assessment. This approach can involve comparing a single topic across multiple models or comparing the entire set of topics generated by a single model. Human evaluators can assess the relevance, interpretability, and overall coherence of the topics; they can provide a domain-specific perspective.^[143] Additionally, qualitative evaluation can also demonstrate the diversity of the entire set of topics generated by a model, offering a comprehensive understanding of the various topics and concepts captured by the model.^[144]

5. Discussion and Challenges

Due to the rise of social media and the Internet, there is now a vast range of text data available, including millions of short documents generated by platforms such as Facebook, Twitter, and Reddit.^[145] These documents can vary greatly in terms of length, quality, volume, vocabulary size, and sparsity. This variability is the reason behind the development of multiple types of topic modeling techniques. It is difficult to determine the most appropriate topic modeling approach that is best suited for a specific data set as there are many models to choose from.^[26] There are different key factors that influence the selection of an appropriate topic model, as shown in **Figure 11**. These factors include data source, dataset size, document length, type of noise, level of noise, preprocessing of data, vocabulary size, and sparsity.^[9] At its core, these factors concentrate on comprehending the characteristics associated with the data. The most crucial factor is the source of the data. To determine a model that generates good topics without further adjustments, it is crucial to examine the dataset being utilized against the datasets that are utilized in the original experiments of that particular model. For instance, if working with a Twitter data set, it would be advantageous to choose a model that has been tested on Twitter data. The size of the data set must also be considered. A model that performs efficiently on smaller data sets may not be capable of processing larger sets containing millions of documents.^[146] Other factors, such as document length, noise levels, sparsity, and document types, also have a significant impact in deciding the optimal model. Despite all the research that has been completed, there are some general conclusions that can be drawn about specific topic modeling methods. LSA is particularly well-suited for applications in which large collections of unstructured text need to be analyzed, such as in the field of text retrieval and information retrieval. LSA is straightforward, computationally efficient, and

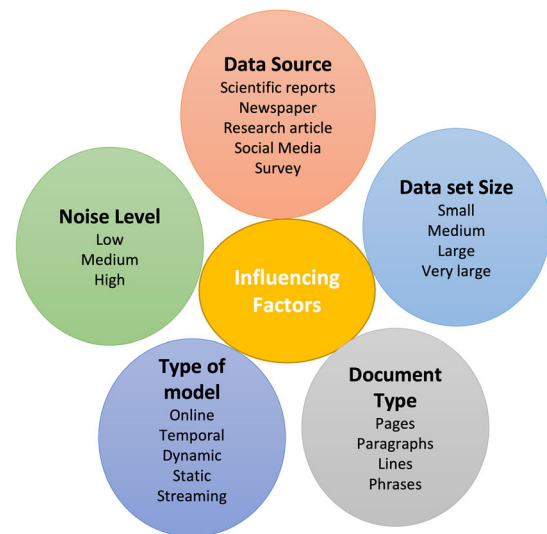


Figure 11. Influencing factors in deciding topic modeling technique.

can handle sparse data well.^[147] pLSA can be a great choice for topic modeling when the goal of the effort involves uncovering the underlying generative process of text data, estimating the probability of generating the observed data, and making predictions about unseen data; however, it may not be the best choice for cases in which computational efficiency is more important or in which the data is very sparse.^[13,148] LDA can be a great choice for topic modeling when the data is large, high-dimensional, and sparse; however, it may not be the best choice for cases in which more fine-grained distinction of topics is needed or in which computational efficiency is more important. Also, LDA can have difficulty with noise filtering.^[31,149] NMF is particularly well-suited when considering applications in which topic interpretability is crucial. Its non-negativity constraints lead to parts-based representations, often yielding more localized and distinct topics. Like LDA and pLSA, NMF uncovers latent structure and facilitates dimensionality reduction. However, it's purely additive, non-negative factorization makes it especially effective in scenarios where negative components lack interpretability. Thus, NMF may be less robust when applied to highly sparse or noisy data.^[150] Similarly, when the data set is large, sparse, and has a large vocabulary with many rare words, the Top2Vec model is very suitable as the embedding technique can handle this kind of data well. BERTopic is a good choice for topic modeling for situations in which the interpretability of the topics is important, in which the data set is sparse, and in which the estimation of the uncertainty of the inferred topics is important. However, it may not be the best choice for cases in which computational efficiency is more important or in which the data is not very diverse.^[109]

Apart from the factors influencing the selection of topic modeling techniques, topic modeling poses several challenges. Here are some of the most important challenges in topic modeling:

5.1. Vocabulary Size

The size of the vocabulary is a major challenge in topic modeling, as the number of unique words can often be quite large, making

it difficult to handle the data efficiently. As the size of the vocabulary increases, the number of possible word combinations also increases, which can cause computational problems.^[151]

5.2. Data Sparsity and Quality

Topic modeling algorithms (e.g., LDA) are in fact designed to handle sparse data by assuming that each document contains only a small subset of all possible topics, and each topic includes a limited set of representative words—this is modeled using Dirichlet priors. However, extreme sparsity, where documents contain very few informative words, or data with many missing values, can still pose challenges in reliably inferring coherent topics. Additionally, the quality of data remains a critical factor. Noisy, unstructured, or incomplete text can hinder the accuracy and interpretability of the generated topics, making preprocessing and data cleaning crucial steps in the topic modeling pipeline.^[152]

5.3. Scalability

As datasets continue to grow in size, topic modeling algorithms often become computationally intensive and increasingly difficult to scale. The runtime complexity typically increases polynomially, depending on the underlying model architecture and inference strategy. Both traditional models like LDA and modern approaches based on deep embeddings can encounter significant scalability challenges when applied to large-scale corpora. To address these limitations, recent systems such as NLP-Fast have been introduced.^[153] These frameworks leverage parallel processing and hardware-aware optimizations to accelerate natural language processing tasks, including topic modeling, thereby achieving substantial improvements in runtime performance on large datasets.

5.4. Overfitting

Some topic modeling algorithms, particularly Bayesian methods like NMF & pLSA, can overfit the training data if it is not properly regularized. This may lead to poor generalization performance on unseen data and result in low-quality or incoherent topics.^[54]

5.5. Interpreting Topics and Human Bias

Topic modeling algorithms often produce topics that are difficult to interpret, as they are represented by a collection of words that have no clear meaning. This can make it difficult to understand the underlying meaning of the inferred topics. Another challenge in topic modeling is the potential for human bias in data collection, in the construction of models, and in the interpretation of the results.^[132] For example, if a topic model is trained on a dataset that contains a disproportionate number of texts written by a certain group of people, the topics that are generated by the model may disproportionately represent the perspectives and experiences of that group.^[154,155]

5.6. Different Types of Text

Topic modeling methods are based on text; however, each type of text has its own characteristics, such as length, density of

information, and language, which can affect the choice of the algorithm, the preprocessing, and the interpretation of the obtained results.^[156]

5.7. Application-Specific Challenges and Solutions

Topic modeling is a powerful approach that is used to discover latent thematic structures in large collections of unstructured text. However, its performance and applicability can vary in significant manner depending on the nature of the application and the characteristics of the data involved. Domains such as short text analysis, specialized technical literature, and multilingual corpora each present distinct challenges that often necessitate customized modeling strategies and domain-aware solutions. In this section, we have summarized the key challenges and corresponding solutions associated with topic modeling across various application domains.

5.8. Healthcare and Bioinformatics

Topic modeling plays an important role in healthcare and bioinformatics through enabling the extraction of meaningful insights from complex, unstructured textual data. In healthcare, it is particularly useful for analyzing clinical narratives from EHRs, supporting tasks such as disease surveillance and diagnostic decision making.^[157] Despite challenges like domain-specific noise and the demand for high interpretability, models like LDA have proven effective in uncovering clinically relevant subtopics, including symptom clusters and drug interactions. In the field of bioinformatics, topic modeling facilitates the summarization of semantic themes across vast, high-dimensional biomedical literature. This domain presents additional difficulties due to data sparsity and the intricacy of biological terminology, often characterized by ambiguous abbreviations and specialized jargon. Top2Vec offers a practical alternative in this context, as it reduces preprocessing requirements and can effectively model large-scale biomedical corpora without requiring a predefined number of topics.^[73]

5.9. Genome-Wide Association Studies (GWAS)

Topic modeling has significantly contributed to advancements in GWAS by uncovering disease subtypes and revealing complex genetic associations. Age-aware models have been particularly impactful, identifying distinct subtypes in conditions such as type 2 diabetes and asthma, thereby improving the accuracy of polygenic risk scores.^[100] Ontology-based approaches like TreeLFA have further enhanced GWAS by uncovering multimorbidity clusters and previously unrecognized cross-disease genetic links.^[101] Traditional models such as LDA and NMF are limited in their ability to capture age-related disease progression and hierarchical relationships among phenotypes. ATM addresses this limitation by incorporating age as a covariate, enabling dynamic modeling of comorbidity evolution over time. Meanwhile, TreeLFA leverages ontology-informed priors to model diseases hierarchically, enhancing both interpretability and relevance for cross-phenotype genetic analyses.

5.10. Social Media Platform

Topic modeling on social media enables the detection of emerging trends, real-time events, and the spread of misinformation by aggregating short, informal texts such as tweets and comments into coherent thematic structures.^[30,158] However, the inherent brevity, linguistic noise, and rapidly evolving nature of social media content pose significant challenges for traditional topic modeling approaches. BERTopic effectively addresses these limitations by leveraging contextualized embeddings, allowing it to capture nuanced, informal, and evolving topics with greater accuracy. This makes it particularly well-suited for dynamic and fast-paced social media environments.

5.11. Scientific Research Articles

Topic modeling in scientific literature facilitates the discovery of latent research themes, improves literature retrieval, and supports topic-based document clustering. However, the high dimensionality, domain-specific jargon, and disciplinary heterogeneity of scientific texts present significant challenges for preprocessing and interpretability. NMF offers a practical solution by producing additive and sparse topic representations, which are often more interpretable and aligned with the conceptual structures familiar to domain experts.^[159]

5.12. News Articles

Topic modeling in news media enables the identification of evolving narratives, detection of bias and agenda framing, and supports large-scale media content monitoring. Key challenges include the presence of overlapping topics within individual articles and the fast-paced nature of news cycles, which can hinder stable topic extraction. Top2Vec provides an effective solution by leveraging semantic embeddings to automatically cluster news articles based on contextual similarity, all with minimal preprocessing. This makes it particularly well-suited for dynamic and time-sensitive news environments.^[73]

5.13. Legal Domain Documents

Topic modeling supports the legal domain by facilitating contract analysis, case law classification, and legal risk identification through topic-based segmentation of dense legal texts. Given the formal structure and complexity of legal language, high interpretability is essential to ensure compliance with legal standards. BERTopic meets this requirement by leveraging transformer-based embeddings to generate granular, context-aware topic representations, making it particularly effective for analyzing intricate legal documents such as briefs and contracts.^[78]

6. Conclusions and Future Work

This survey presents a comprehensive review of the development of some highly popular and most used topic models from their origins to the present day. Topic modeling techniques that have been introduced over time include LSI, NMF, pLSA, LDA,

Top2Vec, and BERTopic. The primary motivation behind the development and continuous evolution of these techniques is the ever-evolving need for researchers to understand their data. Researchers realized the limitations associated with traditional topic models as the volume and characteristics of the data changed over time, which led to the development of more robust topic modeling techniques that can handle noisy, unstructured, and sparse data. This study presents a detailed discussion of the evolution of these techniques, the steps followed in algorithms, as well as the strengths and weaknesses associated with the considered topic models. Generally, these newly introduced models are advanced variants of the old topic modeling techniques. For instance, many LDA variants have been proposed so far to perform topic modeling in different applications, which have performed quite well. Similarly, the NMF introduces the factor matrices for topic representation that adds to the performance of the topic models. Finally, incorporating word embedding vector spaces has introduced new topic models such as Top2Vec and BERTopic, which ultimately lead to further enhancement in the performance associated with topic models on various types of documents.

The initial research revealed exceptional characteristics of the topic modeling techniques, which have generated great interest in various applications, including medicine, healthcare, scientific research, social media, news articles, and legal domains. Some of the current applications have been discussed in this review article. For instance, topic modeling can be utilized with a huge corpus of news articles to identify the latent topics of the articles in newspapers. Similarly, topic modeling has shown exceptional performance in discovering the major concerns of people during the COVID-19 pandemic through Twitter data. The abstracts of the scientific research papers have been analyzed to uncover the latest trends and hot topics of research in the scientific community by applying different topic modeling techniques. Along with an explanation of different domains where topic modeling is applied, a discussion of different evaluation metrics is also provided. Finally, an overview of the factors that influence the selection of topic modeling techniques and the challenges that researchers face while applying these models has been discussed in detail.

This assessment indicates that significant advancements have been made in several areas involving topic modeling in recent years. Several challenges need to be addressed, and goals need to be accomplished in the near future. For instance, most research has focused on static models, and temporal models assume that all data is known in advance. Current topic modeling approaches are not appropriately suited for handling streaming data and changing topics over time. Although online learning has been implemented to improve inference speed to some extent, current models are not designed for handling large streams of temporal data. While earlier topic models struggled with multilingual text, recent advancements, such as BERTopic, have demonstrated the ability to model topics across multiple languages using multilingual transformer-based embeddings. However, there remains a need to further enhance robustness when dealing with highly informal, noisy, or code-switched multilingual text. As text becomes more informal, multilingual, and noisy, future steps related to topic modeling techniques should be optimized to handle these constraints. Moreover, there is still scope

for applying topic modeling in domains such as call center data, environmental sciences, human resources, and educational research. Despite its rich history, there is a need for continued advancements related to topic modeling. For instance, the development of efficient online learning algorithms that can update topic models in real-time as new data comes in can be explored in more detail. Further, most topic models are designed for monolingual data and may not perform well on multilingual datasets, therefore, focused research into multilingual embeddings models that can handle documents in multiple languages effectively is highly desired. Next, a few topic modeling techniques like regularized LSI (regularized LSI), fully sparse topic models (FSTM), and sparse LSI (Sparse LSI) can be ensembled to investigate their effects to reduce noise and handling sparse data more efficiently.

Acknowledgements

P.K. and S.K. contributed equally to this work. The authors would like to thank the Lampe Joint Department of Biomedical Engineering, University of North Carolina at Chapel Hill and North Carolina State University, USA for facilitating the research facilities and desired infrastructure.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

BERTopic, bioinformatics, latent Dirichlet allocation, latent semantic analysis, topic modeling

Received: June 26, 2024

Revised: June 25, 2025

Published online: July 17, 2025

- [1] S. Miles, L. Yao, W. Meng, C. M. Black, Z. B. Miled, *Inf. Process. Manag.* **2022**, 59, 102921.
- [2] Y. Zhang, M. Chen, D. Huang, D. Wu, Y. Li, *Future Gener. Comput. Syst.* **2017**, 66, 30.
- [3] D. J. Newman, S. Block, *J. Am. Soc. Inf. Sci. Technol.* **2006**, 57, 753.
- [4] P. Kumari, D. Toshniwal, *J. Clean. Prod.* **2021**, 279, 123285.
- [5] D. Q. Nguyen, R. Billingsley, L. Du, M. Johnson, *Trans. Assoc. Comput. Linguist.* **2015**, 3, 299.
- [6] M. D. Armstrong, D. Maupomé, M. J. Meurs, Topic Modeling in Embedding Spaces for Depression Assessment. In Canadian AI, **2021**, June 8. The 34th Canadian Conference on Artificial Intelligence.
- [7] P. Nakov, M. A. Hearst, in *Proc. of ACL-08: HLT*, Association for Computational Linguistics, Columbus, Ohio **2008**, pp. 452–460.
- [8] M. R. Ghorab, D. Zhou, A. O'Connor, V. Wade, *User Model User-Adap Inter.* **2013**, 23, 381.
- [9] L. Liu, L. Tang, W. Dong, S. Yao, W. Zhou, *SpringerPlus* **2016**, 5, 1608.
- [10] L. Hong, B. D. Davison, in *Proc. of the First Workshop on Social Media Analytics (SOMA '10)*, Association for Computing Machinery, New York, NY, USA **2010**, pp. 80–88, <https://doi.org/10.1145/1964858.1964870>.
- [11] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, R. Harshman, *J. Am. Soc. Inf. Sci.* **1990**, 41, 391.
- [12] D. Ramage, E. Rosen, J. Chuang, C. D. Manning, D. A. McFarland, *Artif. Intell.* **2009**, 5, 2.
- [13] B. V. Barde, A. M. Bainwad, in *2017 Inter. Conf. on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India **2017**, pp. 745–750, <https://doi.org/10.1109/ICCONS.2017.8250563>.
- [14] Z. Chen, C. Ding, Y. Rao, H. Xie, X. Tao, G. Cheng, F. L. Wang, *World Wide Web.* **2021**, 24, 2139.
- [15] L. Sun, Y. Yin, *Transp. Res., Part C: Emerg. Technol.* **2017**, 77, 49.
- [16] D. M. Blei, J. D. Lafferty, *Ann. Appl. Statist.* **2007**, 1, 17.
- [17] S. Kadian, P. Kumari, S. Shukla, R. Narayan, *Talanta Open* **2023**, 8, 100267.
- [18] O. Koltsova, S. Koltcov, *Policy Inter.* **2013**, 5, 207.
- [19] D. Ramage, S. Dumais, D. Liebling, *Proc. Int. AAAI Conf. Web Soc. Media* **2010**, 4, 130.
- [20] P. Kumari, D. Toshniwal, *Appl. Energy* **2021**, 295, 117061.
- [21] S. Kadian, S. S. Sahoo, P. Kumari, R. J. Narayan, *Electrochim. Acta* **2024**, 475, 143664.
- [22] T.-I. Yang, A. Torget, R. Mihalcea, in *Proc. of the 5th ACL-HLT Workshop on Language Technology for Cultural Heritage, Social Sciences, and Humanities*, Portland, OR, USA, Association for Computational Linguistics **2011**, pp. 96–104.
- [23] S. Sinan Erzurumlu, D. Pachamanova, *Technol. Forecast. Soc. Change* **2020**, 156, 120041.
- [24] P. Kumari, D. Toshniwal, *J. Clean. Prod.* **2021**, 318, 128566.
- [25] K. Rajendra Prasad, M. Mohammed, R. M. Noorullah, *Evol. Intel.* **2021**, 14, 545.
- [26] I. Vayansky, S. A. P. Kumar, *Inf. Syst.* **2020**, 94, 101582.
- [27] T. Hannigan, R. Haans, K. Vakili, H. Tchaljan, V. L. Glaser, M. Wang, S. Kaplan, P. D. Jennings, *Acad. Manag. Ann.* **2019**, 13, <https://doi.org/10.5465/annals.2017.0099>.
- [28] P. Xie, E. P. Xing, Integrating Document Clustering and Topic Modeling. *ArXiv* **2013**, <https://arxiv.org/abs/1309.6874>.
- [29] R. Albalawi, T. H. Yeap, M. Benyoucef, *Artif. Intell.* **2020**, 3, 42.
- [30] X. Cheng, X. Yan, Y. Lan, J. Guo, *IEEE Trans. Knowl. Data Eng.* **2014**, 26, 2928.
- [31] H. Jelodar, Y. Wang, C. Yuan, X. Feng, X. Jiang, Y. Li, L. Zhao, *Multimed. Tools Appl.* **2019**, 78, 15169.
- [32] in *Handbook of Latent Semantic Analysis* (Eds: T. K. Landauer, D. S. McNamara, S. Dennis, W. Kintsch), Lawrence Erlbaum Associates Publishers, Mahwah, NJ **2007**.
- [33] E. Gaussier, C. Goutte, in *Proceedings of the 28th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '05)*, Association for Computing Machinery, New York, NY, USA **2005**, pp. 601–602, <https://doi.org/10.1145/1076034.1076148>.
- [34] T. Hofmann, *Mach. Learn.* **2001**, 42, 177.
- [35] D. Blei, L. Carin, D. Dunson, *IEEE Signal Process. Mag.* **2010**, 27, 55.
- [36] J.-T. Chien, C.-H. Lee, Z.-H. Tan, *Neurocomputing* **2018**, 278, 12.
- [37] X. Li, J. Ouyang, X. Zhou, *Neurocomputing, Part B* **2015**, 149, 811.
- [38] X. Wang, Y. Qiao, Y. Hou, S. Zhang, X. Han, *IEEE Trans. Eng. Manag.* **2021**, 68, 1309.
- [39] B. Karas, S. Qu, Y. Xu, Q. Zhu, *Front. Artif. Intell.* **2022**, 5, 948313.
- [40] M. De Groot, M. Aliannejadi, M. R. Haas, Experiments on Generalizability of BERTopic on Multi-Domain Short Text. *ArXiv* **2022**, <https://arxiv.org/abs/2212.08459>.
- [41] J. Boyd-Graber, Y. Hu, D. Mimno, *Found. Trends. Inf. Ret.* **2017**, 11, 143.
- [42] C. K. Yau, A. Porter, N. Newman, A. Suominen, *Scientometrics* **2014**, 100, 767.
- [43] D. Kuang, P. J. Brantingham, A. L. Bertozzi, *Crime Sci.* **2017**, 6, 12.
- [44] P. Kumari, D. Toshniwal, in *Proc. of Inter. Conf. on Applied Energy, Västerås, Sweden* **2019**, pp. 12–15.
- [45] P. Kherwa, P. Bansal, *EAI Endorsed Trans. Scalable Inf. Syst.* **2020**, 7, e2.

- [46] C. Wang, D. M. Blei, in *Proc. of the 17th ACM SIGKDD Inter. Conf. on Knowledge Discovery and Data Mining (KDD '11)*, Association for Computing Machinery, New York, NY, USA **2011**, pp. 448–456, <https://doi.org/10.1145/2020408.2020480>.
- [47] T. K. Landauer, P. W. Foltz, D. Laham, *Discourse Processes* **1998**, 25, 259.
- [48] S. T. Dumais, G. W. Furnas, T. K. Landauer, S. Deerwester, R. Harshman, in *Proc. of the SIGCHI Conf. on Human Factors in Computing Systems (CHI '88)*, Association for Computing Machinery, New York, NY, USA **1988**, pp. 281–285, <https://doi.org/10.1145/57167.57214>.
- [49] P. Kumari, D. Toshniwal, *Int. J. Green Energy* **2021**, 18, 1007.
- [50] S. Kadian, P. Kumari, S. S. Sahoo, S. Shukla, R. J. Narayan, *Microchem. J.* **2024**, 200, 110350.
- [51] A. Kontostathis, W. M. Pottenger, *Inf. Process. Manag.* **2006**, 42, 56.
- [52] Y. Tonta, H. R. Darvish, *J. Informetr.* **2010**, 4, 166.
- [53] B. T. Bartell, G. W. Cottrell, R. K. Belew, in *Proc. of the 15th Annual Inter. ACM SIGIR Conf. on Research and Development in Information Retrieval (SIGIR '92)*, Association for Computing Machinery, New York, NY, USA **1992**, pp. 161–167, <https://doi.org/10.1145/133160.133191>.
- [54] Y. Lu, Q. Mei, C. Zhai, *Inf. Retr.* **2011**, 14, 178.
- [55] C. X. Zhai, C. Geigle, in *The 41st Inter. ACM SIGIR Conf. on Research and Development in Information Retrieval (SIGIR '18)*, Association for Computing Machinery, New York, NY, USA **2018**, pp. 1395–1398, <https://doi.org/10.1145/3209978.3210189>.
- [56] J. K. Pritchard, M. Stephens, P. Donnelly, *Genetics* **2000**, 155, 945.
- [57] G.-R. Xue, W. Dai, Q. Yang, Y. Yu, in *Proceedings of the 31st Annual International ACM SIGIR Conf. on Research and Development in Information Retrieval (SIGIR '08)*, Association for Computing Machinery, New York, NY, USA **2008**, pp. 627–634, <https://doi.org/10.1145/1390334.1390441>.
- [58] R. Fernandez-Beltran, J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, F. Pla, *IEEE Geosci. Remote Sens. Lett.* **2018**, 15, 1347.
- [59] R. Nallapati, W. Cohen, *Proc. Int. AAAI Conf. Web Soc. Media* **2021**, 2, 84.
- [60] H. Koh, M. Fienup, *Inf. Technol. Libraries* **2021**, 40, <https://doi.org/10.6017/ital.v40i3.13333>.
- [61] H. Patel, D. Kumar Singh, O. Prakash Verma, S. Kadian, in *Inter. Conf. on MAchine Intelligence for Research & Innovations*, Springer Nature Singapore, Singapore **2023**, pp. 227–233.
- [62] D. Blei, A. Ng, M. Jordan, *J. Mach. Learn. Res.* **2001**, 3, 601.
- [63] S. K. Sethi, S. Kadian, G. Manik, *Arch. Comput. Methods Eng.* **2022**, 29, 1.
- [64] A. M. Martinez, A. C. Kak, *IEEE Trans. Pattern Anal. Mach. Intell.* **2001**, 23, 228.
- [65] L.-F. Chen, H.-Y. M. Liao, M.-T. Ko, J.-C. Lin, G.-J. Yu, *Pattern Recognit.* **2000**, 33, 1713.
- [66] J. Yang, J.-Y. Yang, *Pattern Recognit.* **2003**, 36, 563.
- [67] C. Wan, Y. Peng, K. Xiao, X. Liu, T. Jiang, D. Liu, *Inf. Sci.* **2020**, 519, 243.
- [68] W. Xiao-Jun, J. Kittler, Y. Jing-Yu, K. Messer, W. Shitong, in *Proc. of the 17th Inter. Conf. on Pattern Recognition*, 2004, ICPR, Cambridge, UK **2004**, Vol. 4, pp. 545–548, <https://doi.org/10.1109/ICPR.2004.1333830>.
- [69] G. Balikas, M.-R. Amini, M. Clausel, in *Proc. of the 39th Inter. ACM SIGIR Conf. on Research and Development in Information Retrieval (SIGIR '16)*, New York, NY, USA, Association for Computing Machinery **2016**, pp. 921–924, <https://doi.org/10.1145/2911451.2914714>.
- [70] C. Puschmann, T. Scheffler, Topic Modeling for Media and Communication Research: A Short Primer (August 25, 2016). HIIG Discussion Paper Series No. 2016-05 **2016**, <https://ssrn.com/abstract=2836478> or <http://dx.doi.org/10.2139/ssrn.2836478>.
- [71] J. Marjanen, E. Zosa, S. Hengchen, L. Pivovarov, M. Tolonen, Topic modelling discourse dynamics in historical newspapers. *ArXiv* **2020**, <https://arxiv.org/abs/2011.10428>.
- [72] A. Onan, S. Korukoglu, H. Bulut, *Int. J. Comput. Linguistics Appl.* **2016**, 7, 101.
- [73] D. Angelov, Top2Vec: Distributed Representations of Topics. *ArXiv* **2020**, <https://arxiv.org/abs/2008.09470>.
- [74] P. Ma, Q. Zeng-Treitler, S. J. Nelson, in *Int. Conf. ICT Soc. Hum. Beings* **2021**, Vol. 384, pp. 221–226.
- [75] D. Hendry, F. Darari, R. Nurfadillah, G. Khanna, M. Sun, P. C. Condylis, N. Taufik, in *2021 Inter. Conf. on Advanced Computer Science and Information Systems (ICACSIS)*, Depok, Indonesia **2021**, pp. 1–6, <https://doi.org/10.1109/ICACSIS53237.2021.9631322>.
- [76] E. Austin, O. R. Zaiane, C. Largeron, in *Proc. of the 29th Inter. Conf. on Computational Linguistics*, Gyeongju, 12–17 October **2022**, pp. 971–983, <https://aclanthology.org/2022.coling-1.81>.
- [77] F. Zengul, A. Bulut, N. Oner, A. Ahmed, M. Yadav, H. G. Gray, B. Ozaydin, A practical and empirical comparison of three topic modeling methods using a COVID-19 corpus: LSA, LDA, and Top2Vec **2023**. Proceedings of the 56th Hawaii International Conference on System Sciences.
- [78] M. Grootendorst, BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *ArXiv* **2022**, <https://arxiv.org/abs/2203.05794>.
- [79] L. B. Hutama, D. Suhartono, *Informatica* **2022**, 46, 81.
- [80] A. Abuzayed, H. Al-Khalifa, *Procedia Comput. Sci.* **2021**, 189, 191.
- [81] G. Hristova, N. Netov, in *2022 IEEE Inter. Conf. on Big Data (Big Data)*, Osaka, Japan **2022**, pp. 2259–2264, <https://doi.org/10.1109/BigData55660.2022.10020466>.
- [82] E. Jeon, N. Yoon, S. Y. Sohn, *Technol. Forecast. Soc. Change, Part A* **2023**, 186, 122130.
- [83] Y. Woo-Ryeong, Y. Hoe-Chang, *J. Ind. Dev. Business* **2022**, 13, 37.
- [84] F. Alhaj, A. Al-Haj, A. Sharieh, R. Jabri, *Int. J. Adv. Comput. Sci. Appl.* **2022**, 13, <https://doi.org/10.14569/IJACSA.2022.0130199>.
- [85] G. Alexander, M. Bahja, *JMIR Med. Inform.* **2022**, 10, e29385.
- [86] J. Rashid, J. Kim, A. Hussain, U. Naseem, S. Juneja, *BMC Bioinform.* **2022**, 23, 275.
- [87] A. Karami, A. Gangopadhyay, B. Zhou, H. Kharrazi, *Int. J. Fuzzy Syst.* **2018**, 20, 1334.
- [88] J. A. Dawson, C. Kendzierski, Survival-supervised latent Dirichlet allocation models for genomic analysis of time-to-event outcomes. *arXiv preprint arXiv:1202.5999* **2012**, <https://doi.org/10.48550/arXiv.1202.5999>.
- [89] W. Zhao, W. Zou, J. J. Chen, *BMC Bioinform.* **2014**, 15, S11.
- [90] M. J. Paul, M. Dredze, *PLoS One* **2014**, 9, e103408.
- [91] M. Mohammed, R. M. Noorullah, in *Proc. of the Inter. Conf. on Innovative Computing & Communications (ICICC)*, New Delhi, India **2020**.
- [92] A. Baird, Y. Xia, Y. Cheng, *JAMIA Open* **2022**, 5, ooac028.
- [93] C. Meaney, M. Escobar, T. A. Stukel, P. C. Austin, L. Jaakkimainen, *JMIR Med. Inform.* **2022**, 10, e40102.
- [94] S. M. A. Islam, M. Díaz-Gay, Y. Wu, M. Barnes, R. Vangara, E. N. Bergstrom, Y. He, M. Vella, J. Wang, J. W. Teague, A. J. Gruber, C. D. Steele, B. Otl, A. Khandekar, L. B. Alexandrov, *Cell Genom.* **2022**, 2, <https://doi.org/10.1016/j.xgen.2022.100179>.
- [95] M. Pelizzola, R. Laursen, A. Hobolth, *BMC Bioinform.* **2023**, 24, 187.
- [96] H. C. Donker, D. Neijzen, J. de Jong, G. A. Lunter, Multinomial belief networks for healthcare data. *arXiv preprint arXiv:2311.16909* **2023**.
- [97] R. Rosenthal, N. McGranahan, J. Herrero, B. S. Taylor, C. Swanton, *Genome Biol.* **2016**, 17, 31.
- [98] A. Degasperi, T. D. Amarante, J. Czarnecki, S. Shooter, X. Zou, D. Glodzki, S. Morganella, A. S. Nanda, C. Badja, G. Koh, S. E. Momen, *Nat. Cancer* **2020**, 1, 249.

- [99] T. H. McCoy, V. M. Castro, L. A. Snapper, K. L. Hart, R. H. Perlis, *Mol. Med.* **2017**, 23, 285.
- [100] X. Jiang, M. J. Zhang, Y. Zhang, A. Durvasula, M. Inouye, C. Holmes, A. L. Price, G. McVean, *Nat. Genet.* **2023**, 55, 1854.
- [101] Y. Zhang, X. Jiang, A. J. Mentzer, G. McVean, G. Lunter, *Cell Genom.* **2023**, 3, 100371.
- [102] H. S. Choi, W. S. Lee, S. Y. Sohn, *Comput. Secur.* **2017**, 67, 244.
- [103] V. Chakkarwar, S. C. Tamane, in *Smart Innovation, Systems and Technologies*, Springer, Berlin **2020**.
- [104] A. Anantharaman, A. Jadia, C. T. S. Siri, B. N. Adikar, B. Mohan, in *2019 3rd Inter. Conf. on Trends in Electronics and Informatics (ICOEI)*, Tirunelveli, India **2019**, pp. 704–708, <https://doi.org/10.1109/ICOEI.2019.8862599>.
- [105] R. M. Walker, Y. Chandra, J. Zhang, A. van Witteloostuijn, *Public Adm. Rev.* **2019**, 79, 931.
- [106] K. Kukushkin, Y. Ryabov, A. Borovkov, Digital Twins: A Systematic Literature Review Based on Data Analysis and Topic Modeling. *Data* **2022**, 7, 173.
- [107] R. Silveira, C. G. O. Fernandes, J. A. Monteiro Neto, V. Furtado, J. E. P. Filho, in *CEUR Workshop Proc.*, Aachen, Germany **2021**, Vol. 4, pp. 1–10.
- [108] I. Scarpino, C. Zucco, R. Vallelunga, F. Luzzza, M. Cannataro, *BioTech* **2022**, 11, 41.
- [109] D. Atzeni, D. Bacciu, D. Mazzei, G. Prencipe, *Sensors* **2022**, 22, 4925.
- [110] H. Zou, Z. Wei, J. Su, B. Zhao, Y. Xia, N. Zhao, *Secur. Commun. Netw.* **2021**, 2021, 3265300.
- [111] J. Harashima, S. Kurohashi, in *Proc. of the Second Workshop on NLP Challenges in the Information Explosion Era (NLPiX 2010)*, Beijing, China **2010**.
- [112] R. Alshalan, H. Al-Khalifa, D. Alsaheed, H. Al-Baity, *J. Med. Internet. Res.* **2020**, 22, e22609.
- [113] P. Jansson, S. Liu, in *Proc. of the 3rd Workshop on Noisy User-Generated Text*, Copenhagen, Denmark **2017**.
- [114] H. Zankadi, A. Idrissi, N. Daoudi, I. Hilal, *Educ. Inf. Technol.* **2023**, 28, 5567.
- [115] A. Xu, T. Qi, X. Dong, *J. Phys.: Conf. Ser.* **2020**, 1437, 012102.
- [116] M. Asghari, D. Sierra-Sosa, A. S. Elmaghraby, *Inf. Process. Manage.* **2020**, 57, 102340.
- [117] J. A. Lossio-Ventura, S. Gonzales, J. Morzan, H. Alatrasta-Salas, T. Hernandez-Boussard, J. Bian, *Artif. Intell. Med.* **2021**, 117, 102096.
- [118] R. Egger, J. Yu, *Front. Soc.* **2022**, 7, 886498.
- [119] S. K. Ray, A. Ahmad, C. A. Kumar, *Appl. Artif. Intell.* **2019**, 33, 979.
- [120] B. Webber, T. Cohn, Y. He, Y. Liu, in *Proc. of the 2020 Conf. on Empirical Methods in Natural Language Processing (EMNLP)* **2020**.
- [121] J. Mazarura A. de Waal, in *2016 Pattern Recognition Association of South Africa and Robotics and Mechatronics International Conf. (PRASA-RobMech)*, Stellenbosch, South Africa **2016**, pp. 1–6, <https://doi.org/10.1109/RoboMech.2016.7813155>.
- [122] Y. Chen, H. Zhang, R. Liu, Z. Ye, J. Lin, *Knowl.-Based Syst.* **2019**, 163, 1.
- [123] Q. Liu, Z. Zheng, J. Zheng, Q. Chen, G. Liu, S. Chen, B. Chu, H. Zhu, B. Akinwunmi, J. Huang, C. J. Zhang, W. K. Ming, *J. Med. Int. Res.* **2020**, 22, e19118.
- [124] J. O'Neill, C. Robin, L. O'Brien, P. Buitelaar, in *CEUR Workshop Proc.*, Aachen, Germany **2017**, Vol. 3, pp. 4–12.
- [125] R. Viksna, M. Kirikova, D. Kiopa, in *Proc. of the First Inter. Workshop "CAiSE for Legal Documents" (COUrt 2020) co-located with the 32nd Inter. Conf. on Advanced Information Systems Engineering (CAiSE 2020)*, Grenoble, France **2020**, Vol. 2690.
- [126] P. H. L. De Araujo, T. E. de Campos, F. A. Braz, N. C. da Silva, in *Proc. of the Twelfth Language Resources and Evaluation Conf.* **2020**.
- [127] Y. Chen, R. M. Rabbani, A. Gupta, M. J. Zaki, in *2017 IEEE Symp. Series on Computational Intelligence, SSCI 2017 - Proc.*, IEEE, Piscataway, NJ **2018**.
- [128] O. Salaün, F. Gotti, P. Langlais, K. Benyekhlef, in *Frontiers in Artificial Intelligence and Applications*, Vienna, Austria **2022**.
- [129] H. M. Wallach, I. Murray, R. Salakhutdinov, D. Mimno, in *ACM Inter. Conf. Proceeding Series*, Florida, USA **2009**.
- [130] S. W. Thomas, H. Hemmati, A. E. Hassan, D. Blostein, *Empir. Software Eng.* **2014**, 19, 182.
- [131] J. Qiang, Z. Qian, Y. Li, Y. Yuan, X. Wu, *IEEE Trans. Knowled. Data Eng.* **2022**, 34, 1427.
- [132] E. H. Ramirez, R. Brena, D. Magatti, F. Stella, *Neurocomputing* **2012**, 76, 125.
- [133] H. Zhao, L. Du, W. Buntine, in *34th Inter. Conf. on Machine Learning, ICML, Sydney, Australia* **2017**, Vol. 4, pp. 4072–4081.
- [134] M. Hajjem, C. Latiri, *Procedia Comput. Sci.* **2017**, 112, 761.
- [135] D. Maier, A. Waldherr, P. Miltner, G. Wiedemann, A. Niekler, A. Keinert, B. Pfetsch, G. Heyer, U. Reber, T. Häussler, H. Schmid-Petri, *Commun. Methods Meas.* **2018**, 12, <https://doi.org/10.1080/19312458.2018.1430754>.
- [136] D. Mimno, H. M. Wallach, E. Talley, M. Leenders, A. McCallum, in *EMNLP 2011 - Conf. on Empirical Methods in Natural Language Processing, Proc. of the Conf.*, Edinburgh, UK **2011**.
- [137] Y. Fang, L. Si, N. Somasundaram, Z. Yu, in *Proc. of the Fifth ACM International Conf. on Web Search and Data Mining* **2012**.
- [138] S. Liang, Z. Ren, M. De Rijke, in *Proc. of the 37th Inter. ACM SIGIR Conf. on Research & Development in Information Retrieval*, Gold Coast, Queensland **2014**.
- [139] R. Mehrotra, P. Bhattacharya, E. Yilmaz, in *Proc. of the 36th Inter. ACM SIGIR Conf. on Research and Development in Information Retrieval* **2013**.
- [140] D. Newman, S. Karimi, L. Cavedon, in *Proc. of the 14th Australasian Document Computing Symp.*, University of Sydney, Sydney, Australia **2009**.
- [141] H. M. Wallach, I. Murray, R. Salakhutdinov, D. Mimno, in *Proc. of the 26th Inter. Conf. On Machine Learning, ICML 2009*, Montreal, Canada **2009**.
- [142] J. Murdock, C. Allen, in *Proc. of the AAAI Conf. on Artificial Intelligence*, Austin, TX **2015**, Vol. 29, No. 1.
- [143] S. Shukla, J. Jakowski, S. Kadian, R. J. Narayan, *Comput. Struct. Biotechnol. J.* **2023**, 21, 4149.
- [144] Y. Jeong, I. Park, B. Yoon, *Technol. Forecast. Soc. Change* **2019**, 146, 655.
- [145] S. Daenekindt, J. Huisman, *High Educ.* **2020**, 80, 571.
- [146] A. Daud, J. Li, L. Zhou, F. Muhammad, *Front. Comput. Sci. China* **2010**, 4, 280.
- [147] S. H. Mohammed, S. Al-augby, *Indones. J. Electr. Eng. Comput. Sci.* **2020**, 19, 353.
- [148] I. Titov, R. McDonald, in *Proc. of the 17th Inter. Conf. on World Wide Web*, Beijing, China **2008**.
- [149] R. K. Talreja, H. Sable, V. Chaudhary, S. Kadian, M. Singh, M. Kumar, A. Khosla, *ECS Sens. Plus* **2024**, 3, 041602.
- [150] J. Wang, X.-L. Zhang, *Neurocomputing* **2023**, 515, 157.
- [151] Y. Kim, J. H. Kim, J. M. Lee, M. J. Jang, Y. J. Yum, S. Kim, U. Shin, Y. M. Kim, H. J. Joo, S. Song, *Sci. Rep.* **2022**, 12, 13847.
- [152] Y. Chang, L. Kong, K. Jia, Q. Meng, in *2021 IEEE Inter. Conf. on Data Science and Computer Application (ICDSCA)*, Dalian, China **2021**, pp. 294–299, <https://doi.org/10.1109/ICDSCA53499.2021.9650256>.
- [153] J. Kim, S. Hur, E. Lee, S. Lee, J. Kim, in *2021 30th Inter. Conf. on Parallel Architectures and Compilation Techniques (PACT)*, Atlanta, GA, USA **2021**, pp. 75–89, <https://doi.org/10.1109/PACT52795.2021.00013>.
- [154] J. Chang, S. Gerrish, C. Wang, J. Boyd-Graber, D. Blei, *Adv. Neural Inform. Process. Syst.* **2009**, 22, 18.
- [155] Y. Zuo, C. Li, H. Lin, J. Wu, *IEEE Trans. Knowl. Data Eng.* **2023**, 35, 972.

- [156] I. Vulić, W. De Smet, J. Tang, M.-F. Moens, *Inf. Process. Manage.* **2015**, 51, 111.
- [157] Y. Wang, L. Wang, M. Rastegar-Mojarad, S. Moon, F. Shen, N. Afzal, S. Liu, Y. Zeng, S. Mehrabi, S. Sohn, H. Liu, J. *Biomed. Inf.* **2018**, 77, 34.
- [158] W. X. Zhao, J. Jiang, J. Weng, J. He, E. P. Lim, H. Yan, X. Li, in *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, Dublin, Ireland **2011**.
- [159] D. Lee, H. Seung, *Nature* **1999**, 401, 788.



Pratima Kumari is a Data Scientist at Gilead Sciences, based in Raleigh, North Carolina, USA. She brings deep expertise in Artificial Intelligence and Machine Learning, with a strong focus on applying data-driven solutions to complex business and scientific challenges. Before joining Gilead, Dr. Kumari served as a Research Data Scientist at the University of California, San Francisco, where she contributed to interdisciplinary projects at the intersection of healthcare and AI. Her work spans multiple domains of machine learning, including computer vision, natural language processing, and speech recognition, and she has successfully delivered solutions across a variety of real-world applications. Dr. Kumari earned her Ph.D. in Machine Learning and Data Mining from the Department of Computer Science & Engineering at the Indian Institute of Technology (IIT) Roorkee, India. She has published extensively in peer-reviewed international journals and conferences, contributing to the advancement of AI research and its practical applications. Her academic foundation also includes a master's degree in Advanced Computing from MANIT Bhopal and a bachelor's degree in Computer Science & Engineering from KNIT Sultanpur, India.



Sachin Kadian is a Postdoctoral Research Scholar in the Lampe Joint Department of Biomedical Engineering at the University of North Carolina and North Carolina State University, NC, USA. Prior to this, he was a Postdoctoral Research Assistant at Purdue University, USA, and DST, SERB-Overseas Visiting Doctoral Fellow (OVDF) at University of Alberta (UofA), Canada supported by Govt. of India & UofA. He earned his Ph.D. from the department of Polymer and Process Engineering (Major in materials science, nanotechnology, and biomedical engineering) at the Indian Institute of Technology Roorkee, India in 2021. He received his M. Tech. in Biomedical Engineering from NIT Kurukshetra in 2016 and B. Tech. in ECE from the UIET Kurukshetra, India in 2014. He has co-authored over ≈60 publications in reputed SCI journals and international conferences as well as book chapters and patents. Based on his research work, he has received more than 20 awards and recognitions including Outstanding Postdoc Award, Best PhD Thesis Award, Postdoc Travel Grant, Best Paper Awards etc. Currently, he is working on 3D printed wearable microneedles technology to print new customized designs for the development of next generation interstitial fluid sampling, diagnostic, vaccine, drug and cosmeceutical delivery platforms, and next generation personalized healthcare, therapeutic and medical devices.



Mukund Vora is a Data Engineer at CNH Industrial in Phoenix, Arizona, USA. His work focuses on building scalable data pipelines, automating ETL workflows, and optimizing cloud infrastructure with technologies such as AWS, PySpark, and Terraform. He has contributed to the development of data-driven solutions for precision agriculture and autonomous vehicle systems, delivering robust frameworks for telematics and Controller Area Network (CAN) data processing. Mukund previously worked as a Data Science Intern at Raven Industries, where he developed and deployed machine learning models for agricultural automation. He has also held roles in the financial technology sector, developing ESG reporting solutions and workflow automation at MSCI Inc. He holds a Master's degree in Computer Science from North Carolina State University and a Bachelor's degree in Information Technology from the University of Mumbai. His research has been published in multiple peer-reviewed international journals and conferences, contributing to the fields of data engineering, software engineering, and data science.



Pradyumna Vemuri is a Data Engineering Specialist at Gilead Sciences, based in Raleigh, NC, USA. He focuses on designing and delivering scalable data solutions for data lakes, data products, machine learning initiatives, and business intelligence. He holds a Master's degree in Computer Science from North Carolina State University and works on bridging engineering and analytics to support scientific and operational insights. He has also contributed to the research community with a peer-reviewed publication in the domain of Recommendation Systems.



Sumit Kumar is a Professor & Head in the Department of Electronics & Communication, Faculty of Engineering & Technology at Jain (Deemed-to-be University), Bangalore. Prior to joining Jain University, he served as Associate Professor and Head at the Centre for Space Research, under the Division of Research & Development, and Associate Professor in the School of Electronics & Electrical Engineering at Lovely Professional University, Phagwara, Punjab, India. Dr. Kumar earned his Ph.D. from Department of Electronics Engineering at Indian Institute of Technology (IIT-ISM) Dhanbad, India. During his M. Tech. tenure, he was a recipient of the DST-MHRD fellowship and also worked as a Research Trainee at CSIR-NPL, New Delhi during 2010-11. Dr. Kumar completed his B. Tech in Electronics & Communication Engineering from Kurukshetra University, Haryana, India. He possesses over 17 years of research and academic experience across several reputed institutions in India, including CEN Laboratories, IIT Bombay; IIT (ISM), Dhanbad; CSIR-NPL, New Delhi; the Division of Research & Development, LPU Punjab; and the Nano-Bioelectronics Lab, DTU New Delhi. His research interests include Nanoelectronics, Optoelectronics, Biosensors, and applications of Artificial Intelligence in various electronic domains. He has published a substantial number of high-impact research articles in reputed journals, along with several publications through renowned conferences, book chapters, and editorials. He is also a member of prestigious professional organizations such as IEEE, ISTE, IAENG, and IETE.



Om Prakash Verma is an Assistant Professor in the Department of Instrumentation and Control Engineering at Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar. He received his Ph.D. from IIT Roorkee and M.Tech from NIT Jalandhar. His academic journey reflects a strong foundation in advanced control systems, soft computing, machine vision, and quantum learning. Dr. Verma has authored over 100 research publications in reputed SCI/Scopus-indexed journals, conferences, and book chapters, and has filed multiple patents. He has guided two Ph.D. scholars to completion and is currently supervising four more in the areas of intelligent systems and UAV navigation. He has successfully led and contributed to six funded research projects as PI and Co-PI, supported by prestigious agencies including ISRO, MeitY, and CSIR. His project on acoustic signal-based drone detection, funded by ISRO, received special appreciation for timely and quality completion. He serves as a Senior Member of IEEE and is actively associated with several professional societies such as the IEEE Computational Intelligence Society, IEEE Control Systems Society, and the Instrument Society of India. Dr. Verma also serves as an Associate Editor for the International Journal of Security and Privacy in Pervasive Computing, and frequently chairs sessions at national and international conferences.



Roger J. Narayan is a Distinguished Professor in the Joint Department of Biomedical Engineering at the University of North Carolina and North Carolina State University. He is an author of over two hundred publications as well as several book chapters on the processing of biomedical materials. He currently serves as an editorial board member for several academic journals, including as associate editor of Applied Physics Reviews (AIP Publishing). Dr. Narayan has also edited several books, including the textbook Biomedical Materials, Second Edition (Springer), the handbook Materials for Medical Devices (ASM International), and the Encyclopedia of Biomedical Engineering (Elsevier).